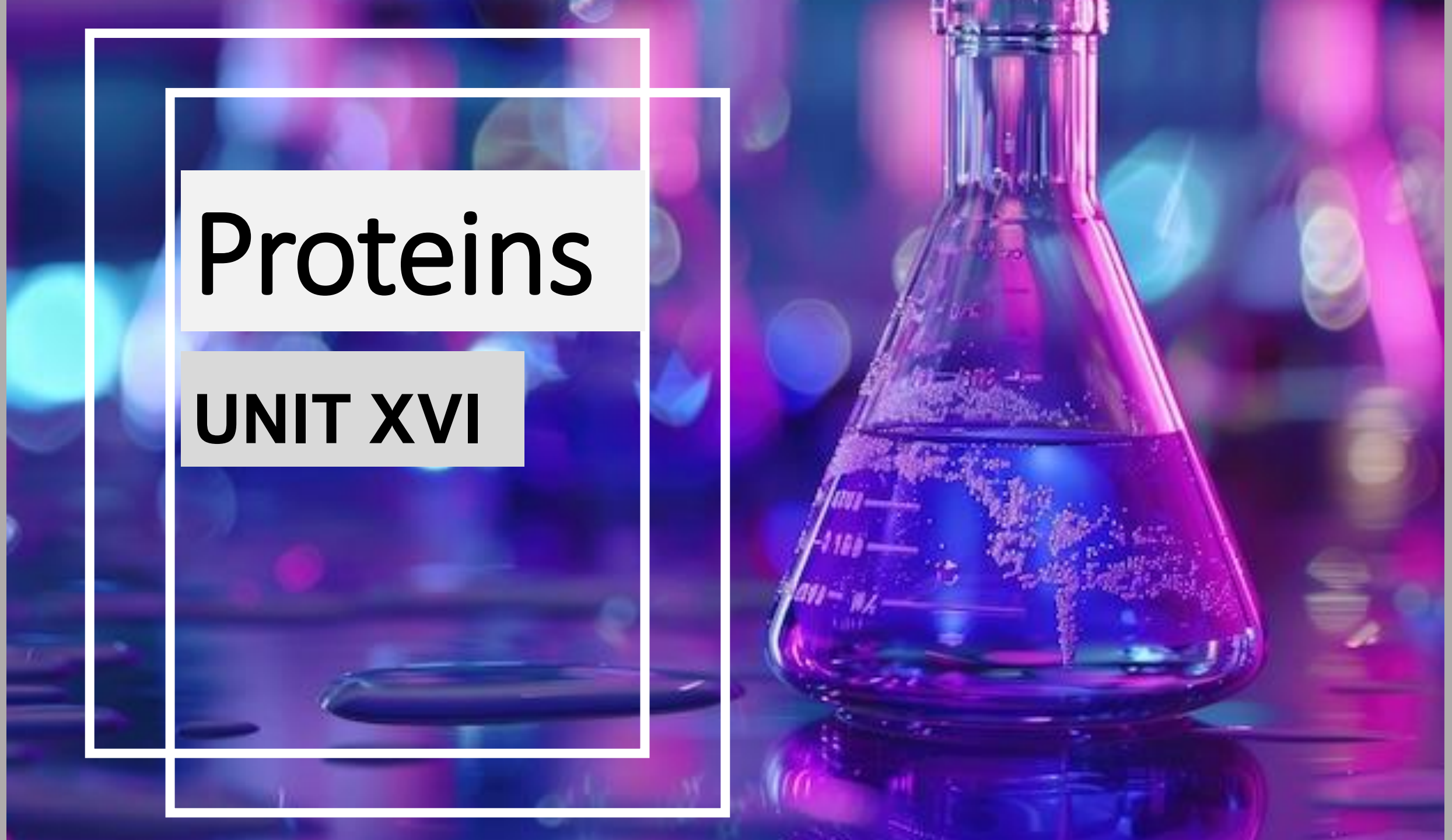


Proteins

UNIT XVI

Learning Outcomes

At the end of each topic, the students should be able to:

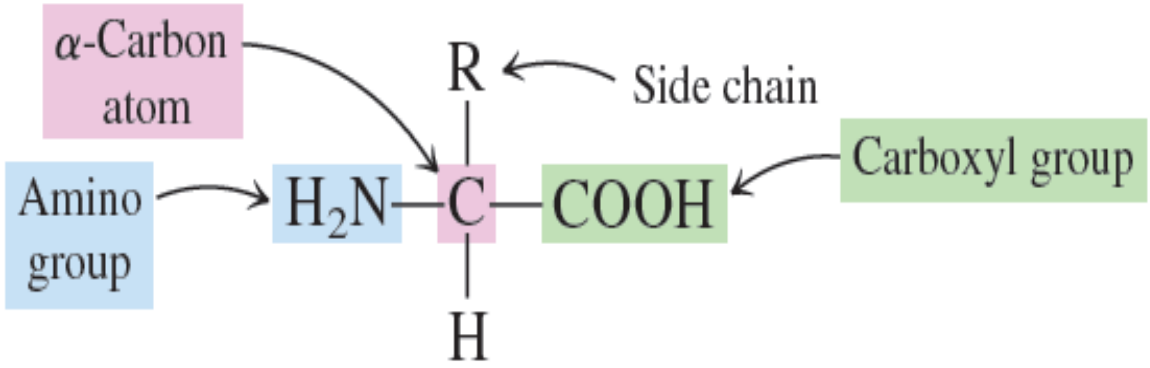
- ❑ Characterize a protein in terms of subunits present in its polymeric structure and elements present
- ❑ Draw the generalized structure for an α-amino acid and classify them into four categories based on structure and polarity of side chains
- ❑ Be familiar with essential amino acids, complete, incomplete and complementary dietary proteins & limiting amino acids
- ❑ Know the handedness pattern for standard amino acids; designated using Fischer projection
- ❑ Understand the four structures of protein: Primary, Secondary, Tertiary & Quaternary and how is it specified, common types, attractive forces and requirements respectively

PROTEINS

- Naturally-occurring, unbranched polymer in which the monomer units are amino acids
- Most abundant substance in cells after water
  - *Account for about 15% of a cell's overall mass*
- Elemental composition - Carbon (C), hydrogen (H), nitrogen (N), oxygen (O), and sulfur (S)
  - *Average nitrogen content is 15.4% by mass*
- Contain iron (Fe), phosphorus (P), and other metals in certain specialized proteins

Amino Acids

- Contain both an amino ($—NH_2$) and a carboxyl ($—COOH$) group
  - **α-amino acids**: Amino acids in which the amino group and the carboxyl group are attached to the α-carbon atom
    - Side chains (R) - Vary in size, shape, charge, acidity, functional groups present, hydrogen-bonding ability, and chemical reactivity
  - >700 amino acids are known



Standard Amino Acids

- 20 α-amino acids normally found in proteins
- Divided based on the properties of R-groups
  - **Nonpolar amino acids**: Contain one amino group, one carboxyl group, and a nonpolar side chain
    - Hydrophobic - Not attracted to water molecules
    - Found in the interior of proteins, where there is no polarity
  - **Polar amino acids** - Hydrophilic
    - Types - Polar neutral, polar acidic, and polar basic

Amino Acids

TABLE 3.1 Names and Abbreviations of the Common Amino Acids

Amino Acid	Three-Letter Abbreviation	One-Letter Abbreviation
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

Note: One-letter abbreviations start with the same letter as the name of the amino acid where this is possible. When the names of several amino acids start with the same letter, phonetic names (occasionally facetious ones) are used, such as Rginine, asparDic, Fenylalanine, tWyptophan. Where two or more amino acids start with the same letter, it is the smallest one whose one-letter abbreviation matches its first letter.

Amino Acids-Building Blocks of Proteins

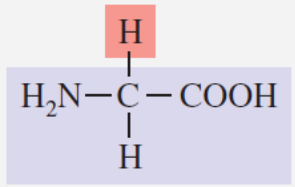
Four Categories according to side chain polarity

1. **Nonpolar amino acid**

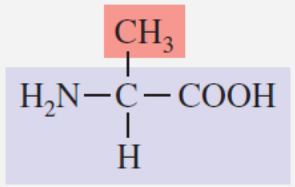
   - has nonpolar side chains – usually hydrocarbons
   - *Hydrophobic*
   - usually hide inside the folded structure of protein
   - Nine nonpolar amino acids

Table 20.1 - 20 Standard Amino Acids

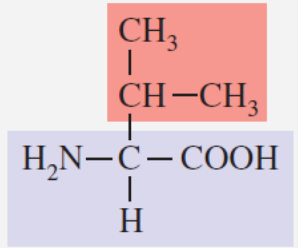
Nonpolar Amino Acids



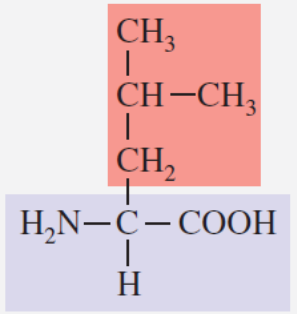
Glycine (Gly, G)
GLY-seen



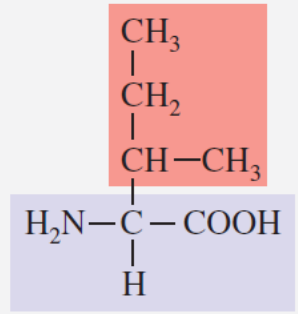
Alanine (Ala, A)
AL-ah-neen



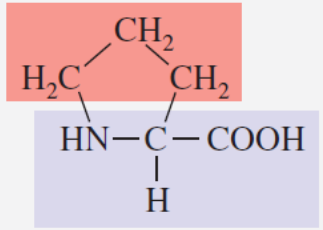
Valine (Val, V)
VAY-leen



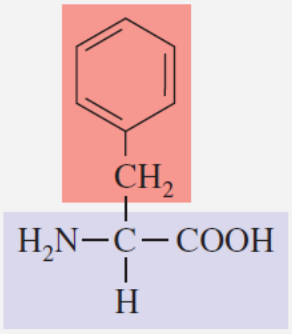
Leucine (Leu, L)
LOO-seen



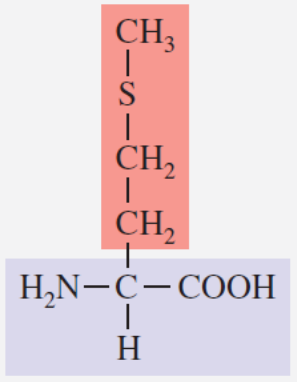
Isoleucine (Ile, I)
eye-so-LOO-seen



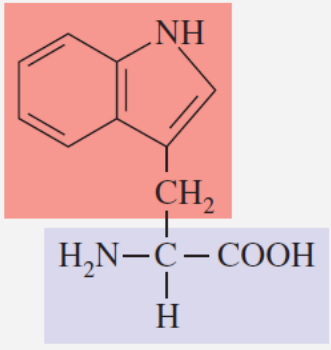
Proline (Pro, P)
PRO-leen



Phenylalanine (Phe, F)
fen-il-AL-ah-neen



Methionine (Met, M)
me-THIGH-oh-neen



Tryptophan (Trp, W)
TRIP-toe-fane

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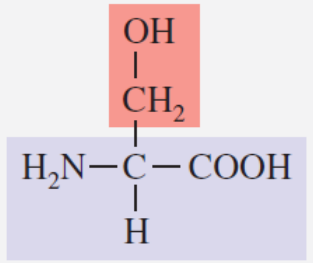
Amino Acids-Building Blocks of Proteins

Four Categories according to side chain polarity

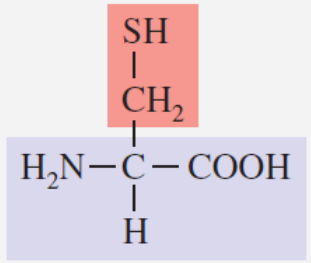
2. **Polar neutral amino acid**:
   - polar but neutral side chain
   - physiological pH neither basic or acidic.
   - 6 amino acids.
   - More soluble in water, R group can H bond to H_2O

Table 20.1 - 20 Standard Amino Acids

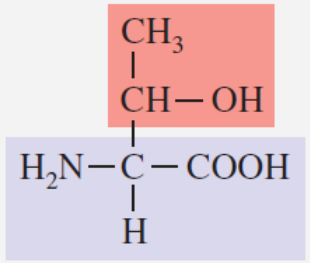
Polar Neutral Amino Acids



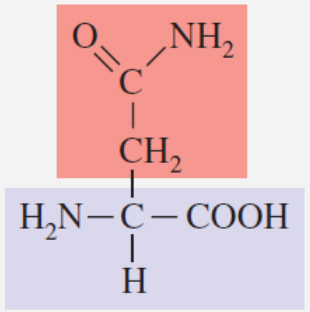
Serine (Ser, S)
SEER-een



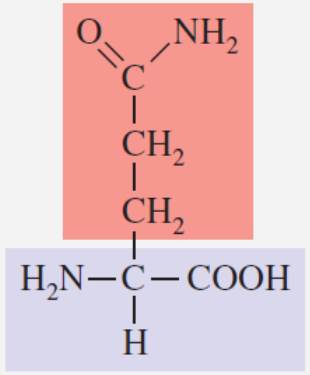
Cysteine (Cys, C)
SIS-teh-een



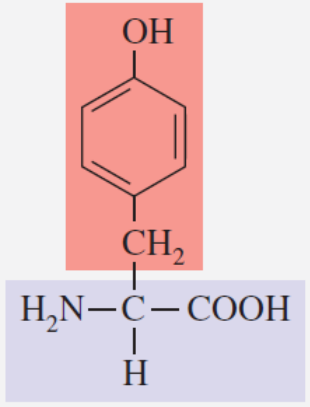
Threonine (Thr, T)
THREE-oh-neen



Asparagine (Asn, N)
ah-SPAR-ah-geen



Glutamine (Gln, Q)
GLU-tah-meen

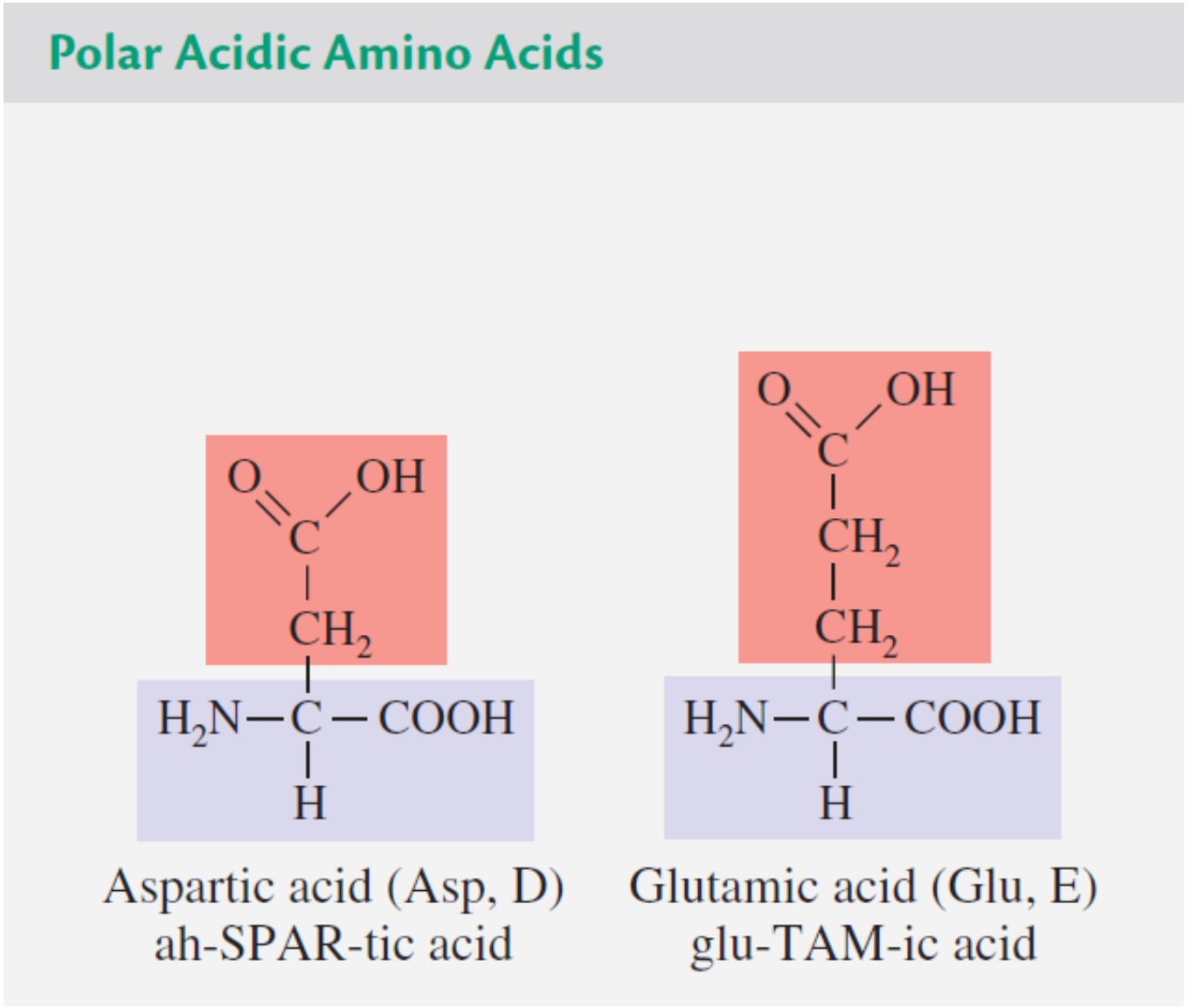


Tyrosine (Tyr, Y)
(TIE-roe-seen)

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3. **Polar acidic amino acid**

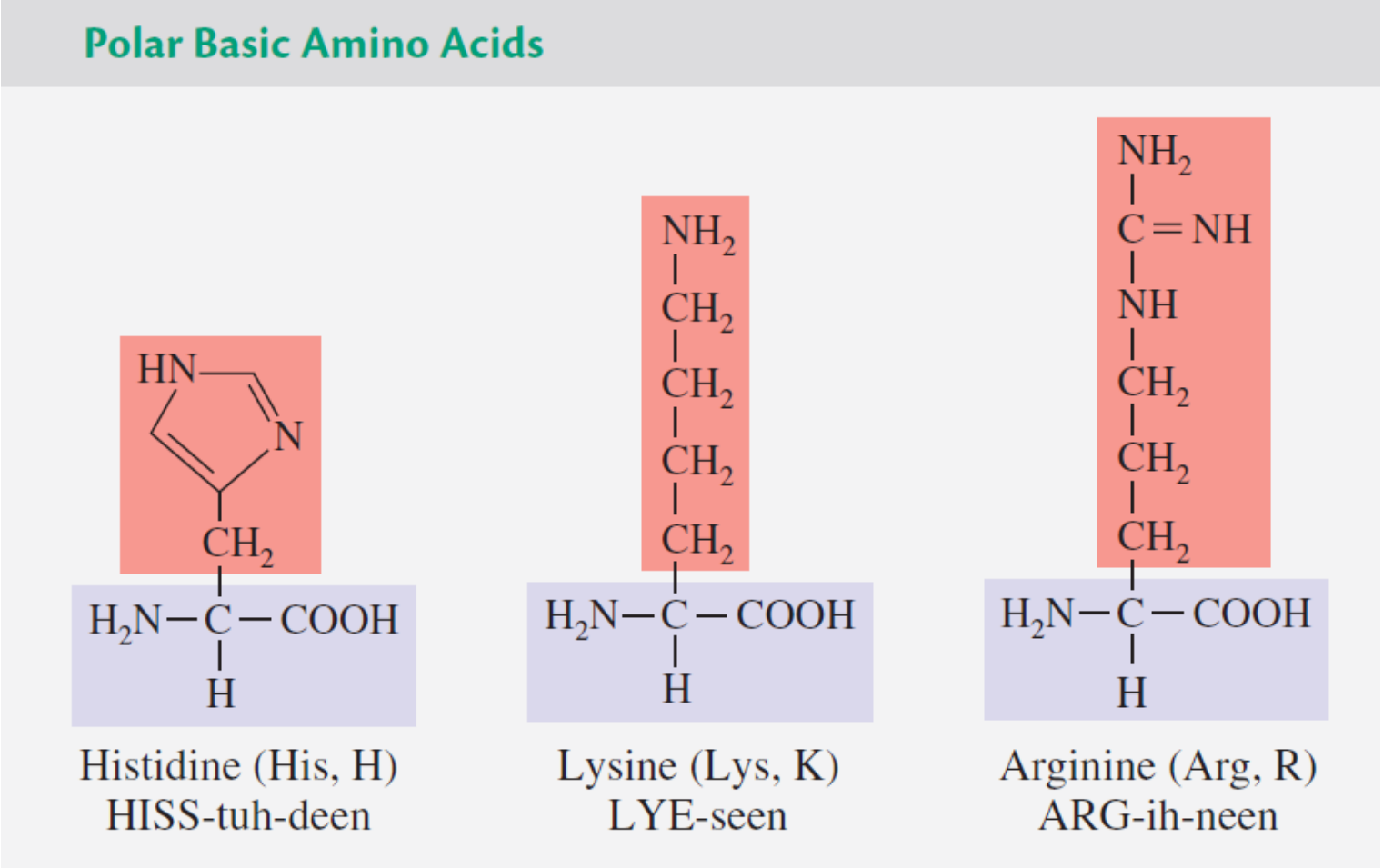
- (-COOH) group part of the side chain
- Two amino acids belong to this category



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4. **Polar basic amino acid**

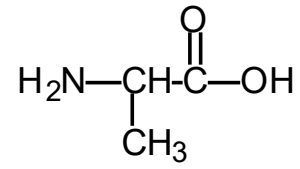
- Contain an amino group as part of the side chain
- Three amino acids belong to this category



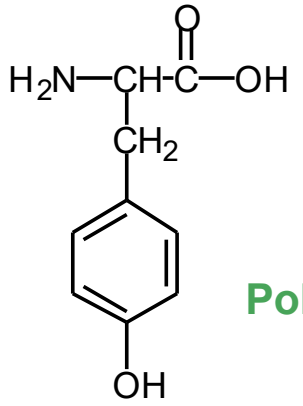
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Exercise

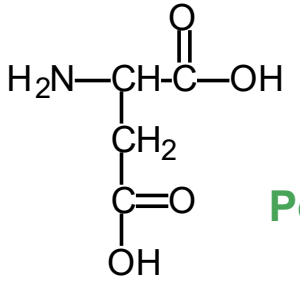
Classify the following amino acids based on the polarity of their R-groups.

a. 

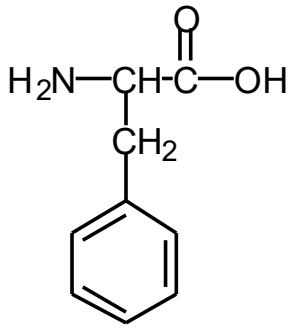
Nonpolar

b. 

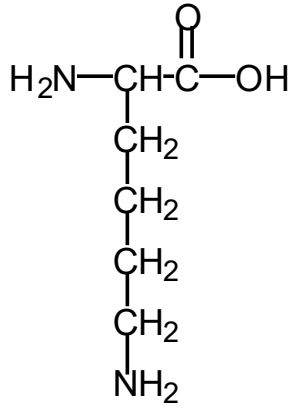
Polar Neutral

c. 

Polar Acidic

d. 

Nonpolar

e. 

Polar Basic

Chirality of Amino Acids (Optical Isomers)

- Four different groups attached to the a-carbon atom in all of the standard amino acids **except glycine**.
- Proteins and Amino Acids are L-isomers by nature

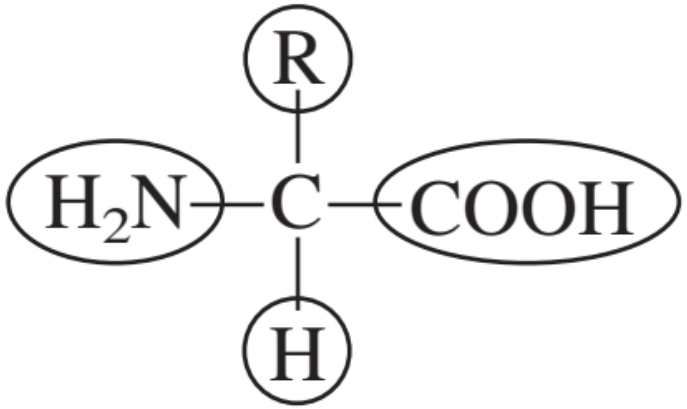


Fig. 2 Chiral structure of amino acid

Chirality of Amino Acids (Optical Isomers)

- Fischer Projection Rules
  a. -COOH put at **top**; R group at **bottom**
  b. -NH_2 at horizontal; left = L; Right = D

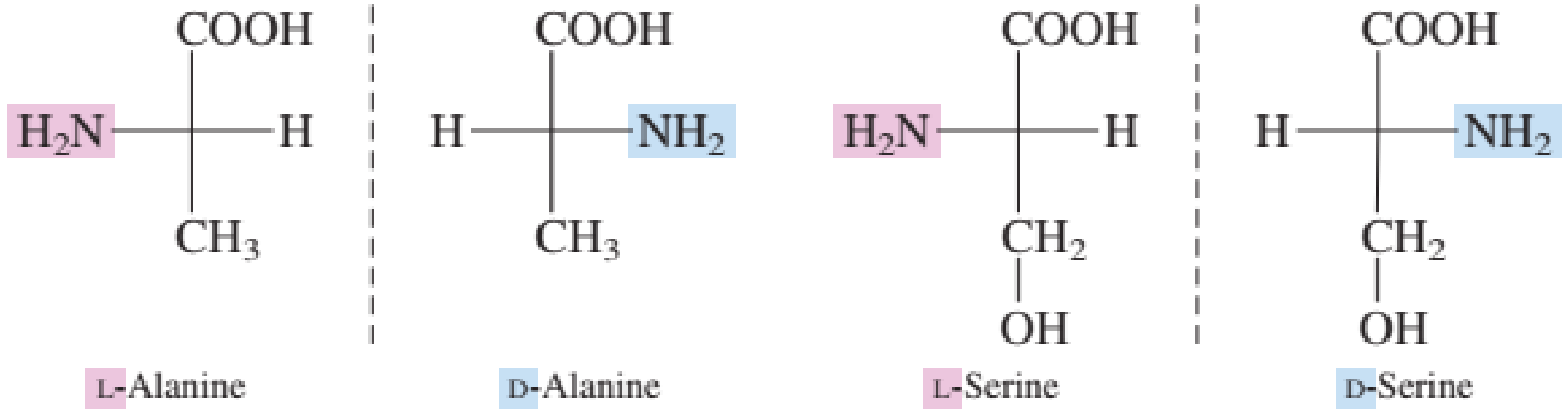
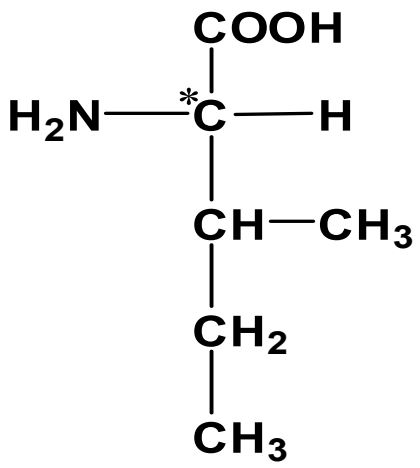


Fig. 6 Fischer projection formulas for both enantiomers of the amino acids alanine and serine

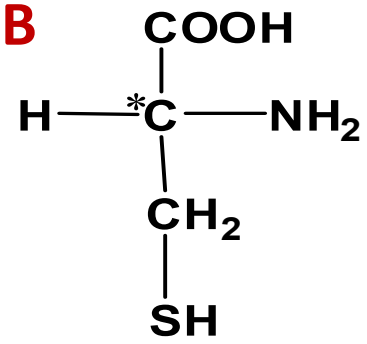
Exercise

Name the following amino acids with correct designation for the enantiomer (chiral carbon is indicated by *).

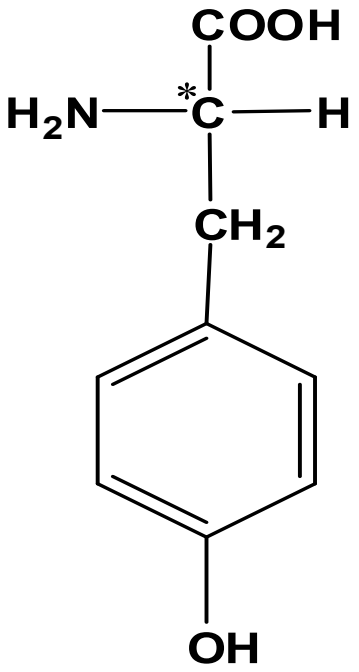
A



B



C



A = L-isoleucine

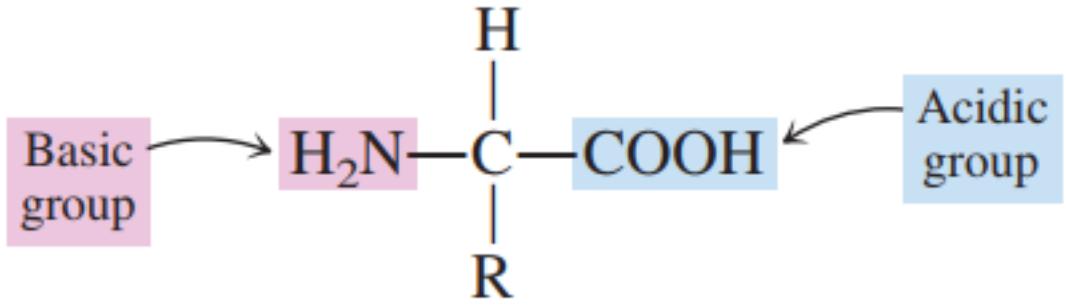
B = **D-cysteine**

C = L-tyrosine

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Acid- Base Properties of Amino Acids

- Both an acidic group (-COOH) and a basic group (-NH_2) are present on the same carbon in an a-amino acid.



- in neutral solution, carboxyl groups have a tendency to lose protons (H+)

$$—COOH \longrightarrow —COO^- + H^+$$

- in neutral solution, amino groups have a tendency to accept protons (H+)

$$—NH_2 + H^+ \longrightarrow —\overset{+}{N}H_3$$

Acid- Base Properties of Amino Acids

- Zwitterion - molecule that has a (+) charge on one atom & (-) charge on another atom, but which has no net charge
- Three species are in equilibrium with each other, and the equilibrium shifts with pH change

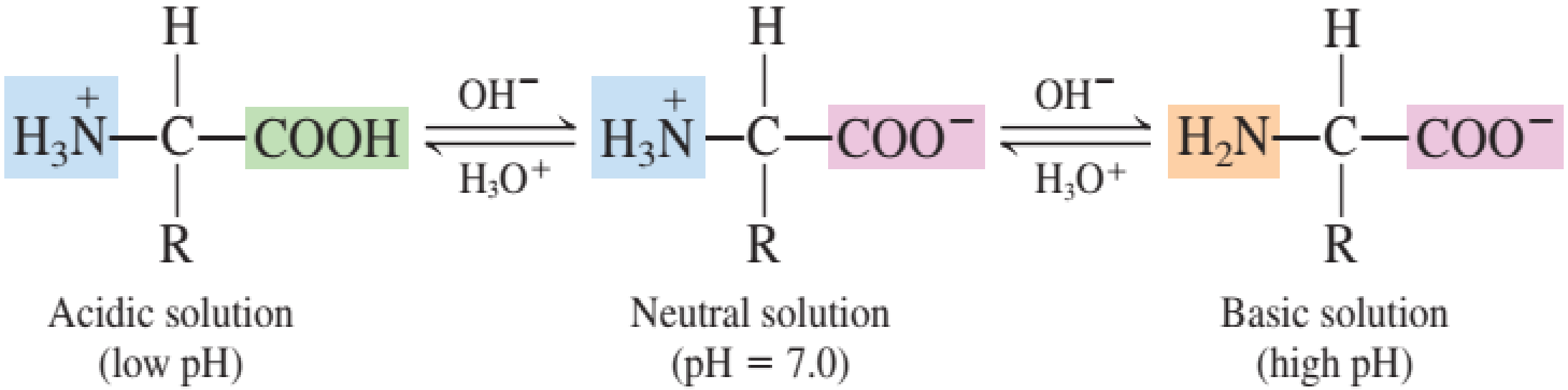


Fig. 7 Three amino acid forms that exist in solution

Acid- Base Properties of Amino Acids

Isoelectric Points pH at which an amino acid exists primarily in its zwitterion form

- 15 amino acids w/ nonpolar or polar neutral side chains @ pH4.8-6.3
- 3 basic amino acids - higher isoelectric points, & two acidic have lower ones

Table 2. Essential amino acids

alanine	6.01	leucine	5.98
arginine	10.76	lysine	9.74
asparagine	5.41	methionine	5.74
aspartic acid	2.77	phenylalanine	5.48
cysteine	5.07	proline	6.48
glutamic acid	3.22	serine	5.68
glutamine	5.65	threonine	5.87
glycine	5.97	tryptophan	5.88
histidine	7.59	tyrosine	5.66
isoleucine	6.02	valine	5.97

When pH < pI, there is an excess amount of H^+ in solution. The excess H^+ is attracted to the negatively charged carboxylate ion resulting in its protonation

When pH > pI, there is an excess amount of OH^- in solution. The excess OH^- is attracted to the positively charged amine group resulting in the removal of an H^+ ion

Cysteine - Chemically Unique Amino Acid

- Contain sulfhydryl group (-SH group)
- *Dimerizes* with another cysteine to form cystine
- Two cysteine residues linked by **disulfide bond**

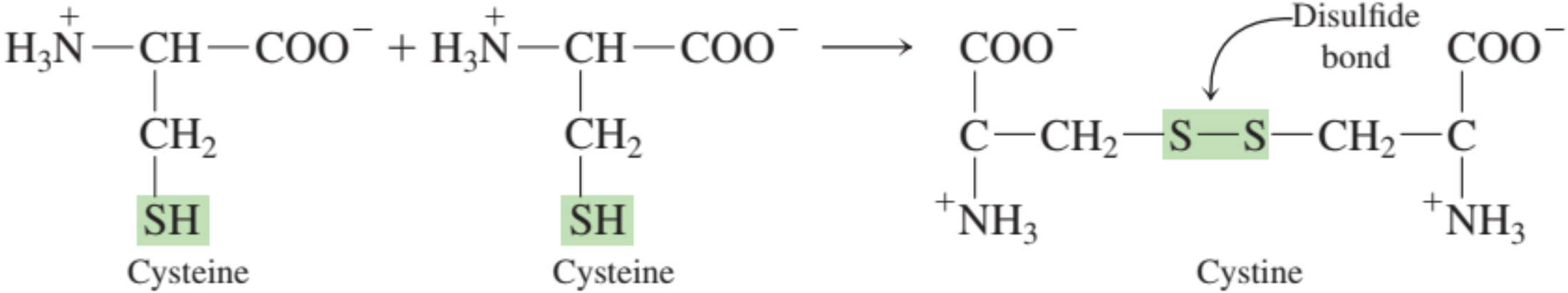


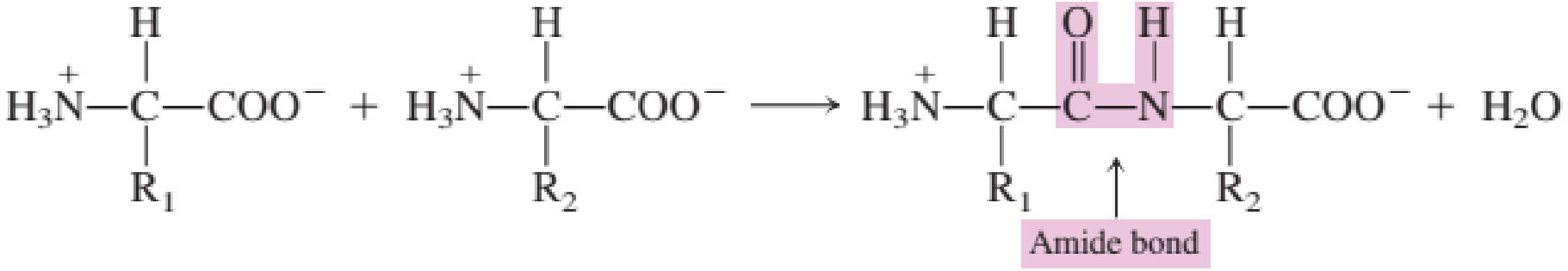
Fig. 8 *Cysteine residues are linked via a covalent disulfide bond*



Fig. 8.1 *oxidation–reduction behavior involving sulfhydryl groups and disulfide bonds*

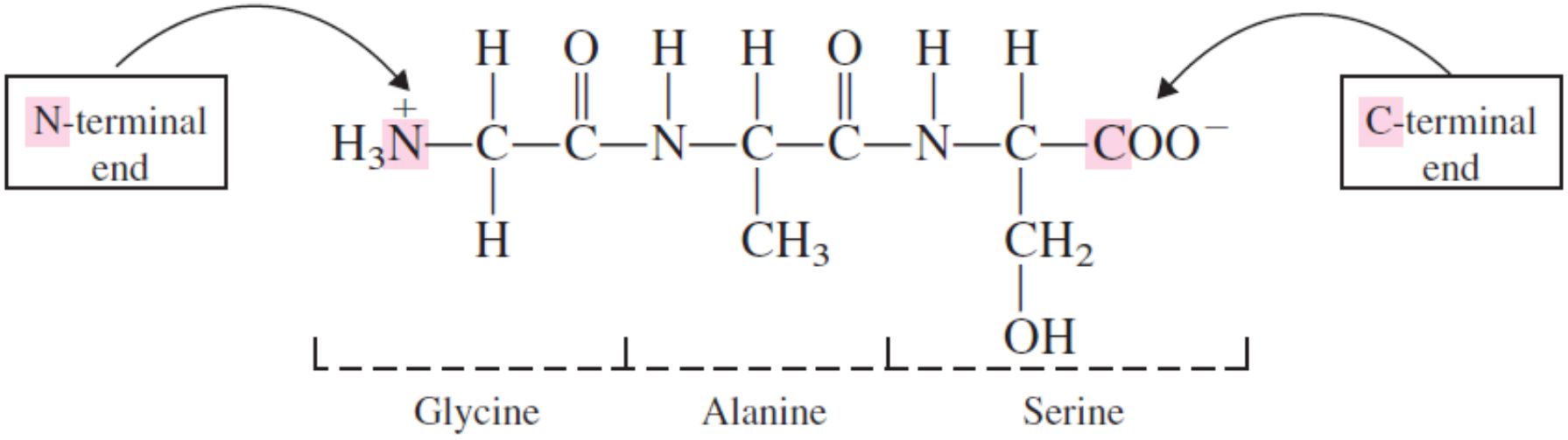
Peptides to Proteins

- amino acids can bond together to produce a peptide chain
  - **Peptide**: Unbranched chain of amino acids
    - **Dipeptide** - Compound containing two amino acids
    - **Oligopeptide** - Peptide with 10 to 20 amino acid residues
    - **Polypeptide**: Long unbranched chain of amino acids
  - Reaction is between the amino group of one amino acid and the carboxyl group of another amino acid



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- Length of the amino acid chain can vary from a few amino acids to hundreds of amino acids
  - **Peptide bonds**: Covalent bonds between amino acids in a peptide
- Every peptide has an N-terminal end and a C-terminal end



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LEVELS of the Structure of Proteins

Primary Structure of Proteins

Primary Protein Structure - order in which amino acid are linked together in proteins

- Order of attachment in peptide bonds
- Same regardless where protein is found in organism
- Peptide bonds = backbone of protein
- C and N atoms arranged in zigzag

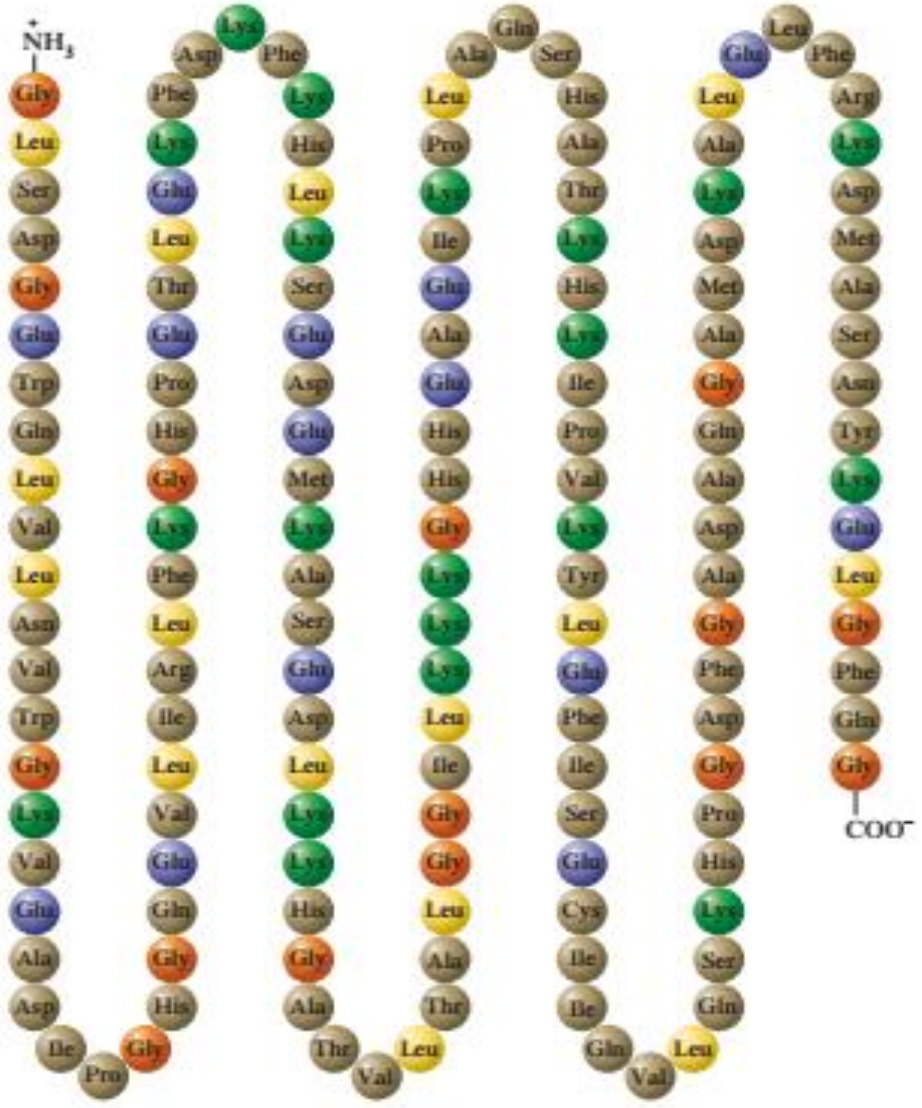
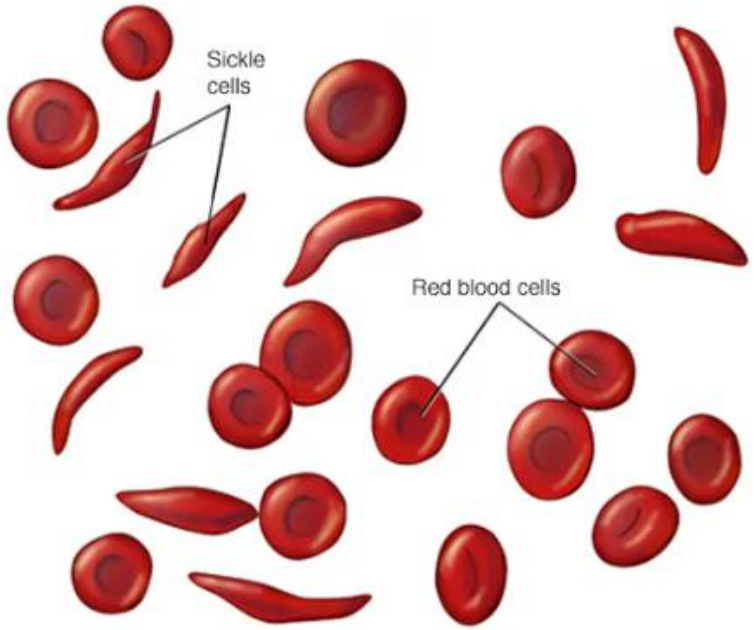


Fig. 12 Primary structure of human myoglobin (not the actual shape)

Sickle Cell Anemia

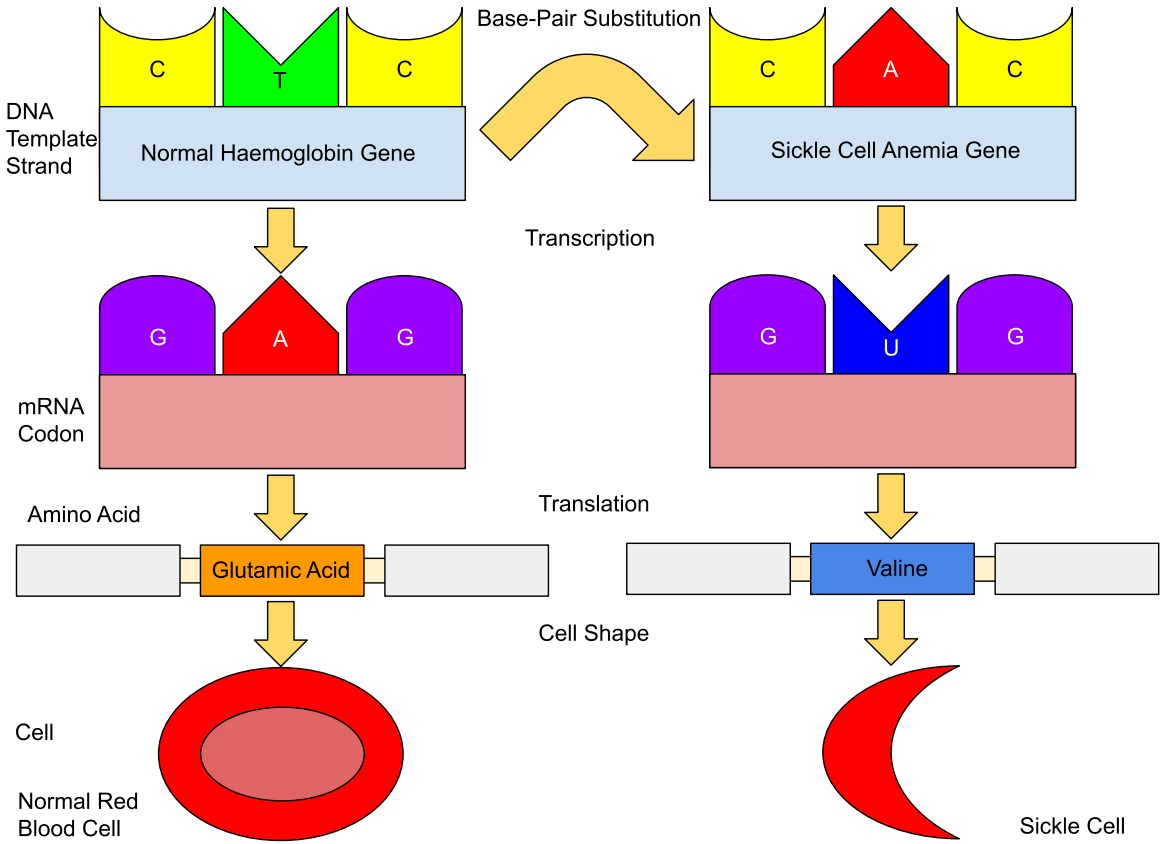
The hemoglobin associated with sickle cell anemia causes red blood cells to become rigid, sticky and misshapen.



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The gene defect is a mutation in **glutamate being substituted by valine at position 6** of the β-globin chain

1  Specifying the primary structure of a protein involves specifying the:

a. Number of peptide linkages present

b. Number of amino acids presents

c. The number and sequence of amino acids present

d. The number of peptide linkages and the amino acids present

Primary Structure of Proteins

Peptide linkages can be planar or zigzag

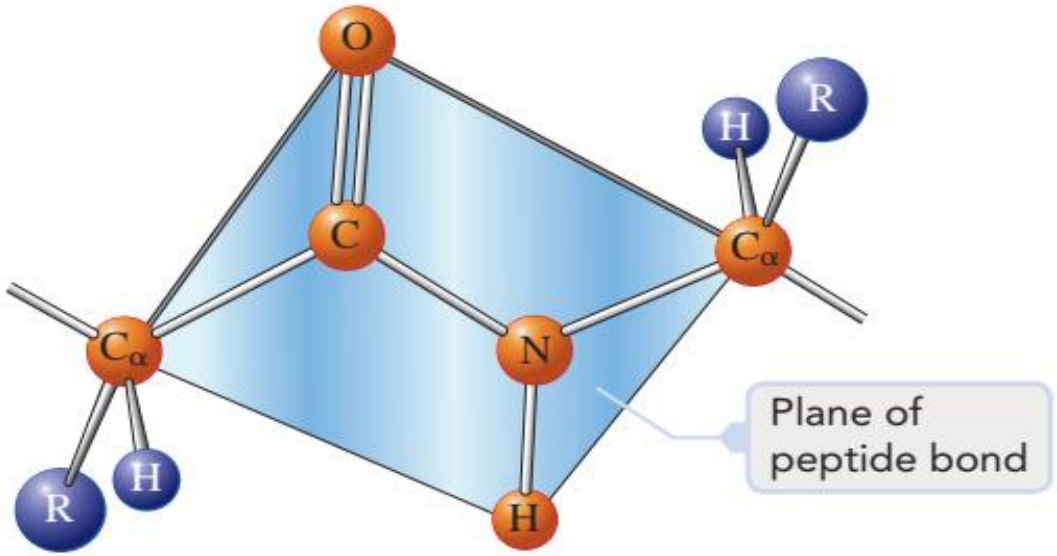


Fig. 13 Six atoms lie in plane of peptide bond

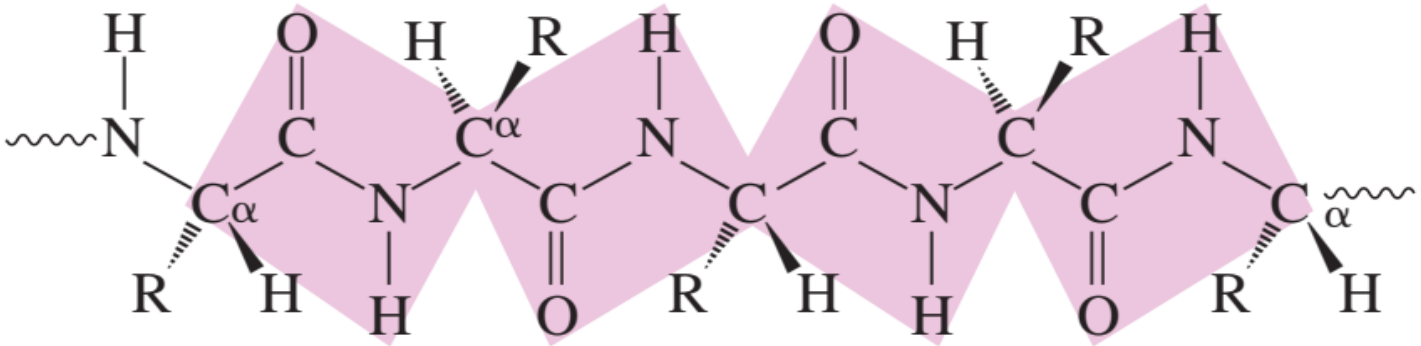
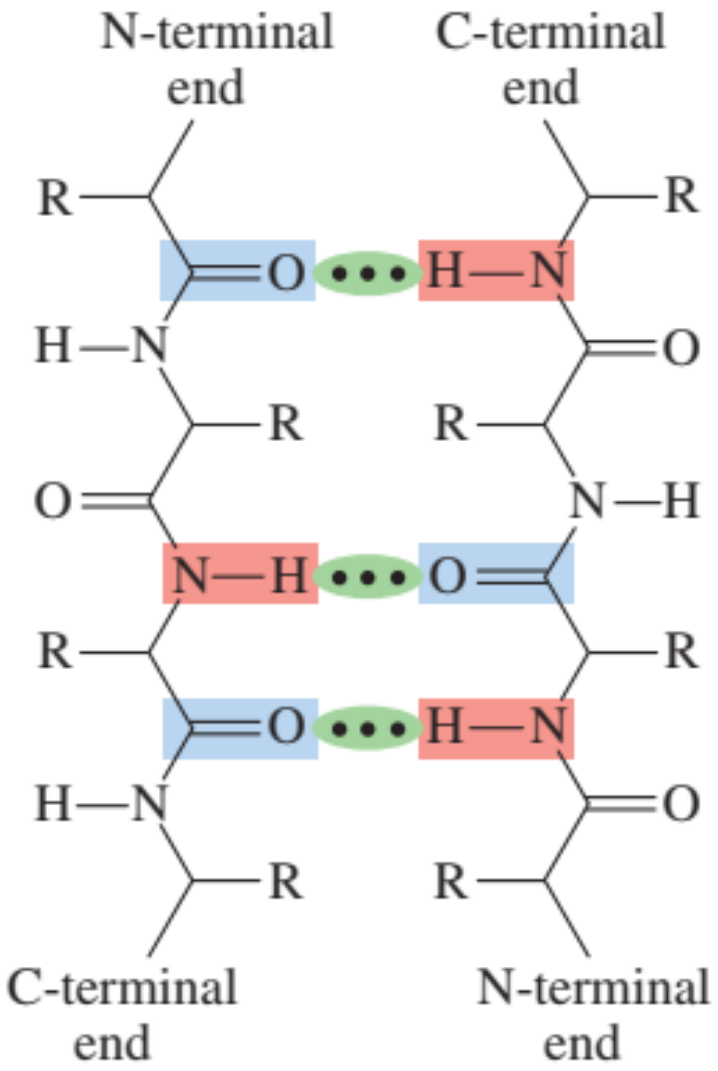


Fig. 14 Zigzag arrangement within protein backbone

Secondary Structure of Proteins



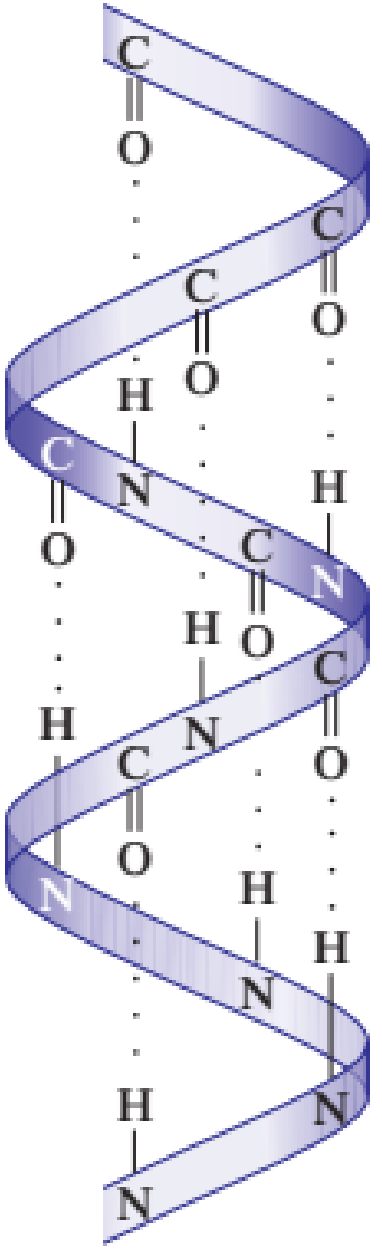
Arrangement in space adopted by backbone portion of protein (α-helix), (β-pleated sheet)
Hydrogen bonding between Carbonyl oxygen atom of peptide linkage and Hydrogen atom of amino group
H bond can be two segments from diff. backbone or two segments of same backbone that folded back upon itself

Fig. 15 hydrogen bonding between carbonyl oxygen atom of one peptide linkage and the amide hydrogen atom of another peptide linkage

Secondary Structure of Proteins

The α-helix - protein chain = coiled spring; maintained by H bonds (intramolecular)

1. Twist - right handed, clockwise or spiral
2. H bond between C=O and N-H are parallel to axis of helix
3. H bond involves C=O and N-H group of amino acid four amino acid residues along the spiral; 1 turn of spiral is = 3.6 amino acid residue
4. Amino acid R groups extend outward from spiral, no room for R group within



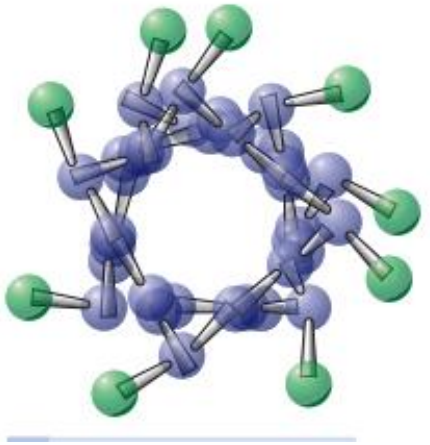


Fig. 16 & 16.1 Top view of helix showing that R groups point away & H bonding in backbone arrangement

Secondary Structure of Proteins

The (β-pleated sheet) - two extended protein segments in same or different molecules held together by H bonds

- Either in single chain that folds back to itself (intrachain), or atoms of different peptide chain (interchain)
- In molecules where β-pleated sheet involves single molecule, U-turns are needed to form structure

Pleated Sheet - zigzag structure

1. H bond between C=O and N-H lie in plane of the sheet
2. Amino Acid R group found on top or bottom, above or below plane of sheet

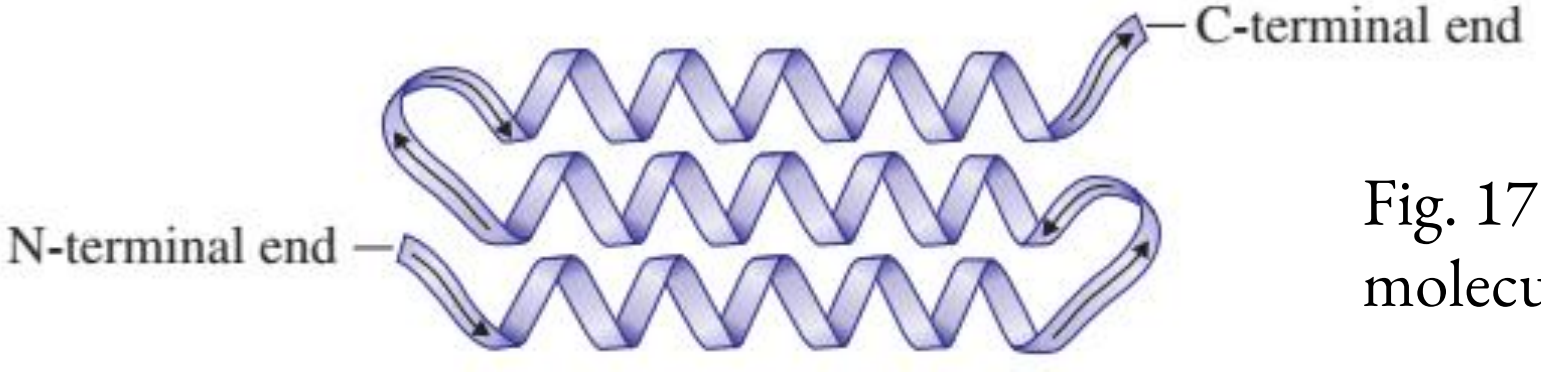
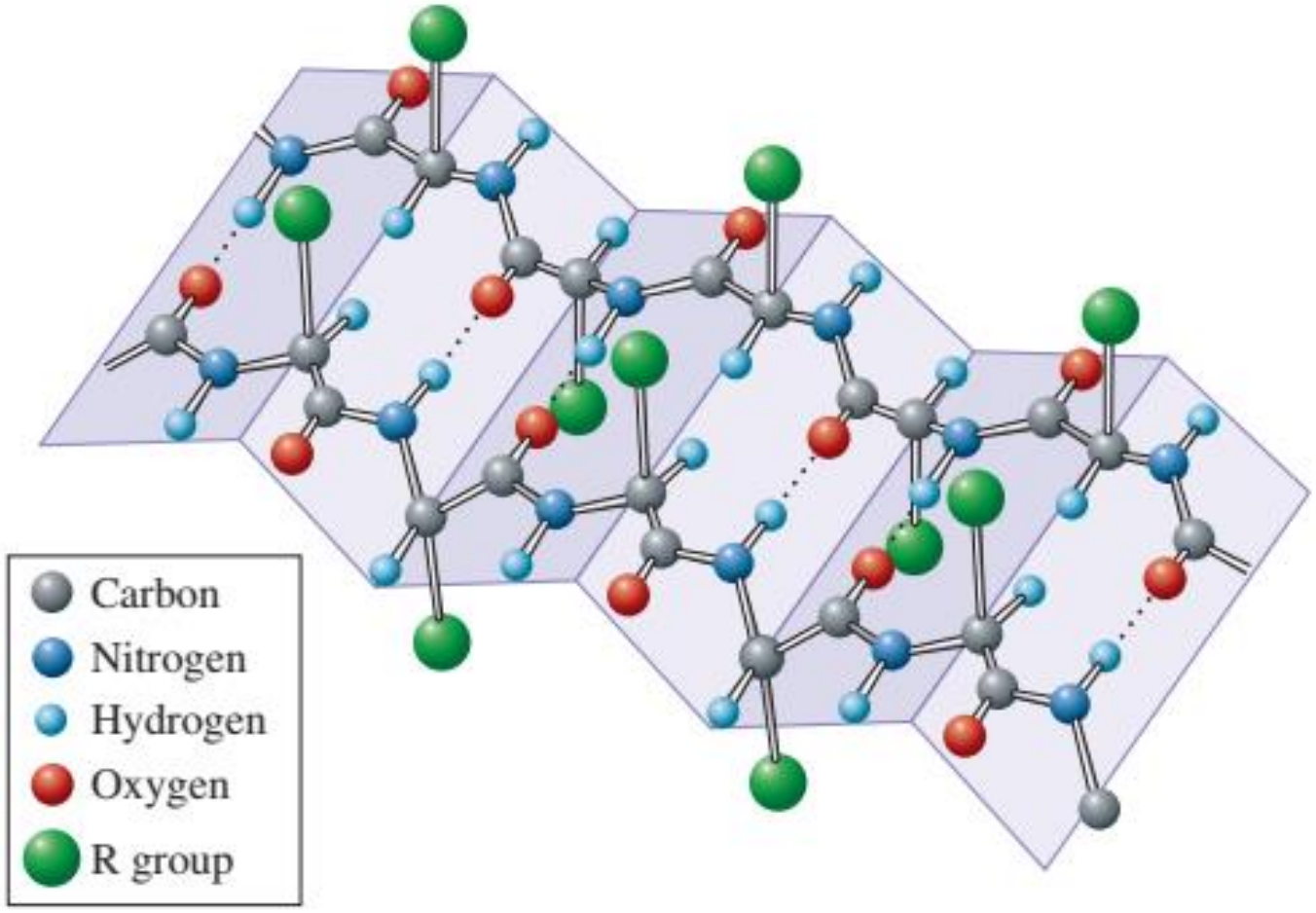
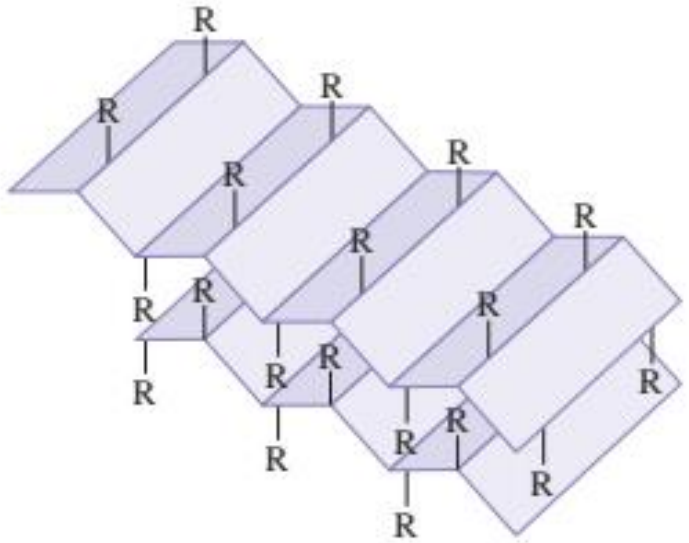


Fig. 17 U-turn in protein chain of single molecule

Secondary Structure of Proteins



a A representation emphasizing the hydrogen bonds between protein chains.



b A representation emphasizing the pleats and the location of the R groups.

Fig. 18 Two representations of β-pleated sheet protein structure

Secondary Structure of Proteins

Unstructured segments - contain neither (α-helix) or (β-pleated sheet)

- few proteins have secondary structures but possible to have both;

- (α-helix) or (β-pleated sheet) found only in protein portions where amino acid R groups present are small;

- large R groups tend to disrupt both of these types of secondary structure

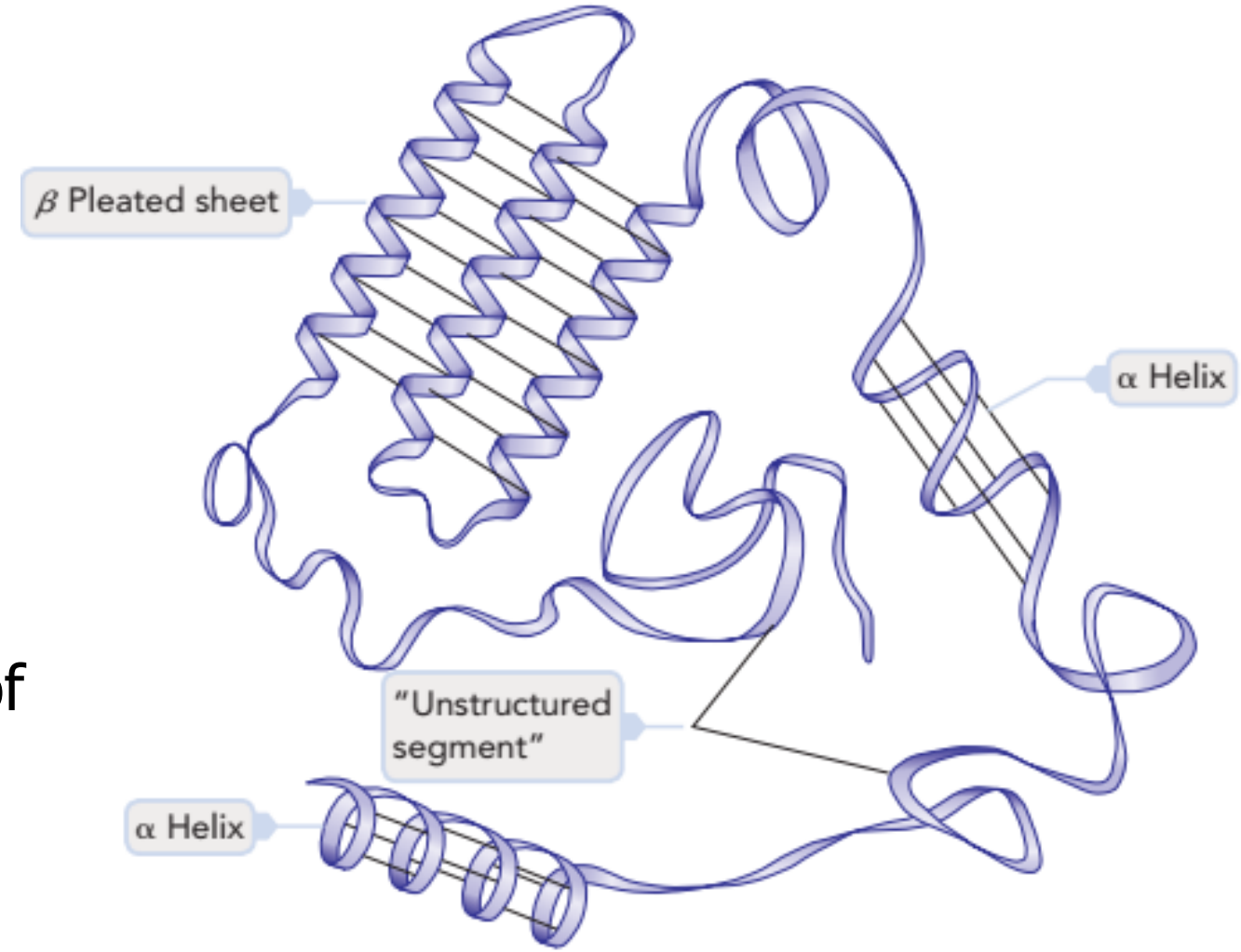


Fig. 19 pleated sheet configurations, as well as areas of "unstructure.

2 Which of the following types of interactions is responsible for protein secondary structure?



a. Hydrogen bonding between C=O and N-H entities

b. Hydrogen bonding between R-groups

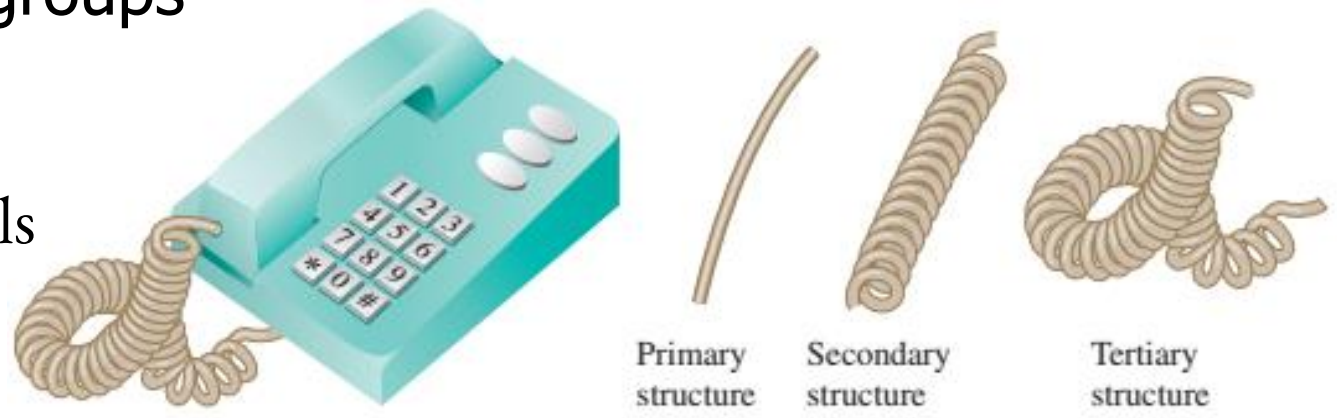
c. Peptide bond between amino acids

d. All intermolecular forces within the chain

Tertiary Structure of Proteins

-3D shape from interaction of R groups

Fig. 20 analogy for the first three levels of protein structure



Interactions Responsible for Tertiary Structure

1. Disulfide Bonds - strongest bond for tertiary structure,-SH of two cysteine to form covalent bond ; intramolecular or intermolecular disulfide bond
2. Electrostatic Interaction (Salt Bridges) - acidic R group and basic R group: at different pH carry charges -COO- and NH3+, cation and anion
3. Hydrogen Bonds - occur in amino acids with polar R groups capable of H-bonding; weak, disrupted by pH changes

Tertiary Structure of Proteins

Interactions Responsible for Tertiary Structure

4. Hydrophobic interactions - two nonpolar side chains, attractive force is *London dispersion forces*, usually between alkyl and phenyl, weaker than H bonds or electrostatic but significant due to number

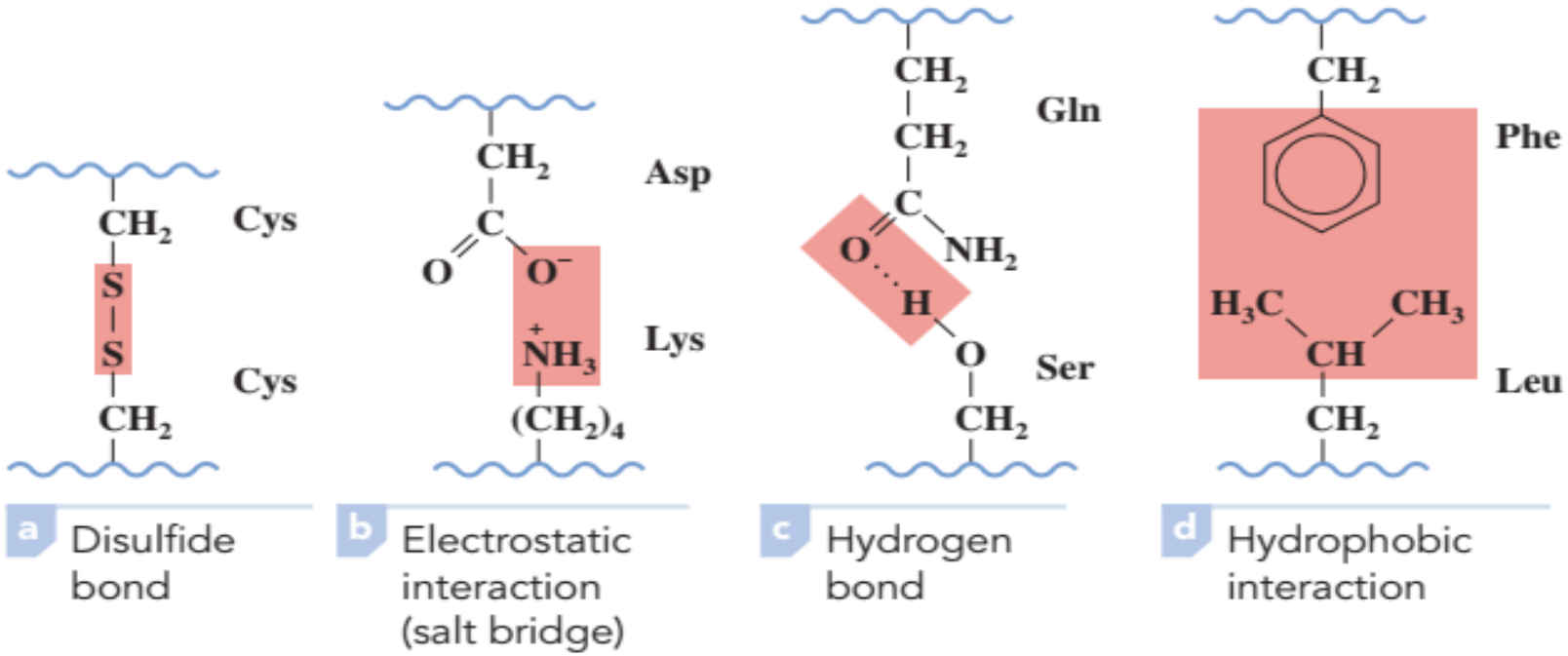


Fig. 21 four types of stabilizing interactions between amino acid R groups that contribute to the tertiary structure of a protein.

Fig. 22 four stabilizing influences that contribute to the tertiary structure of a protein.

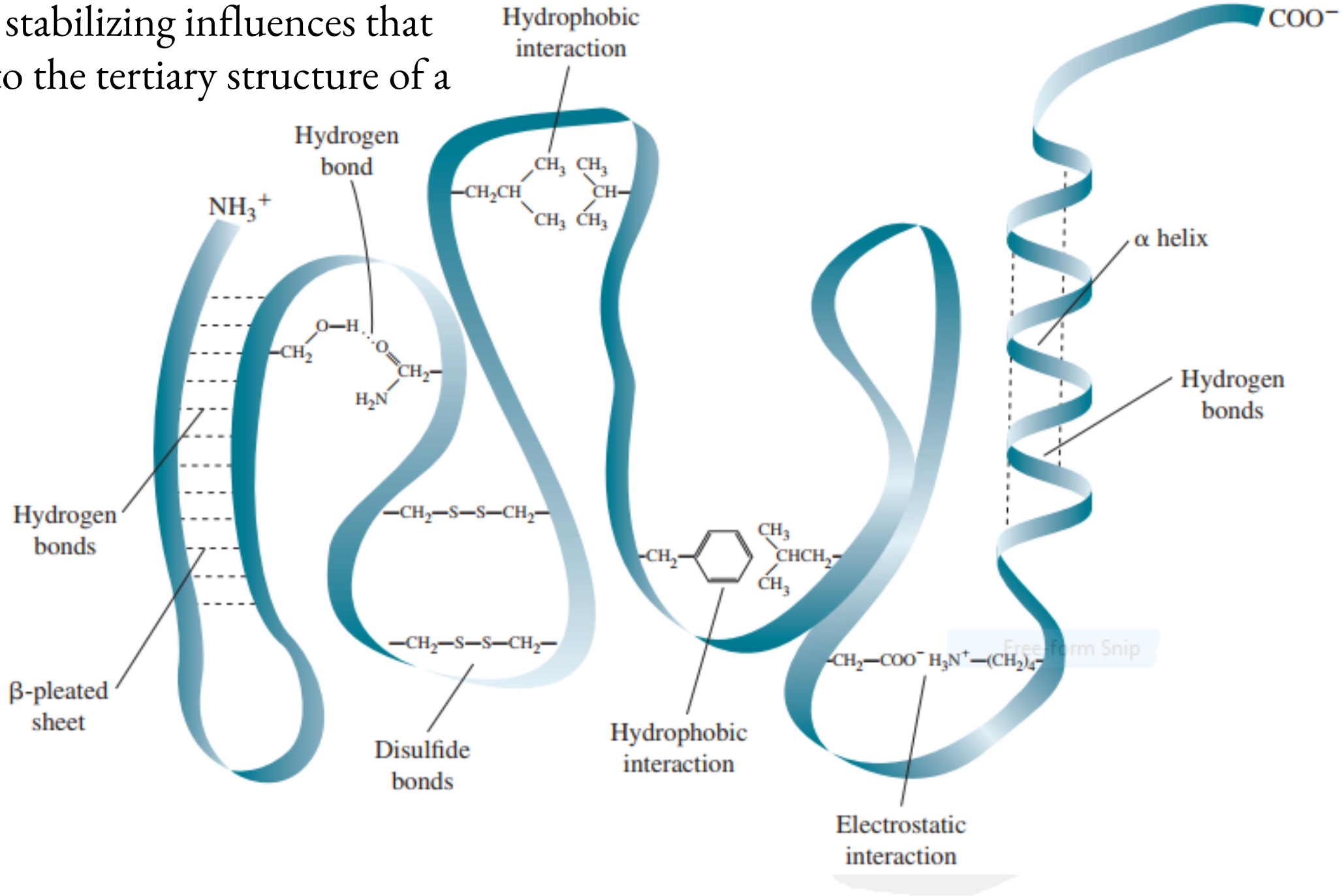


Figure 4.14 (C) - Tertiary Structure of α-Lactalbumin

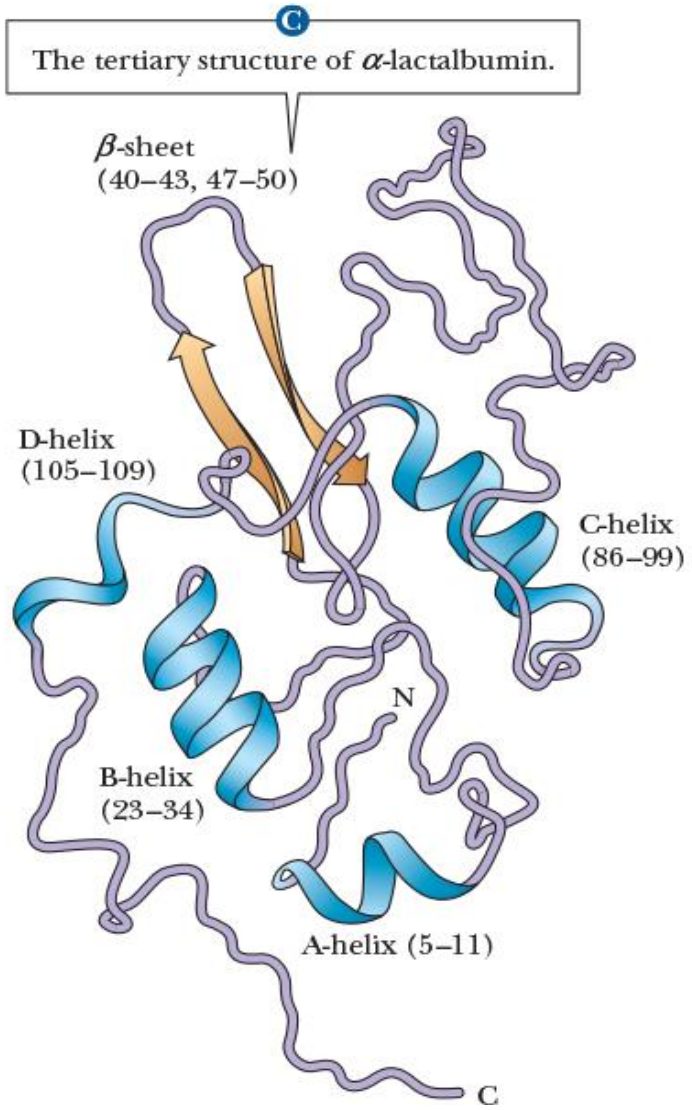
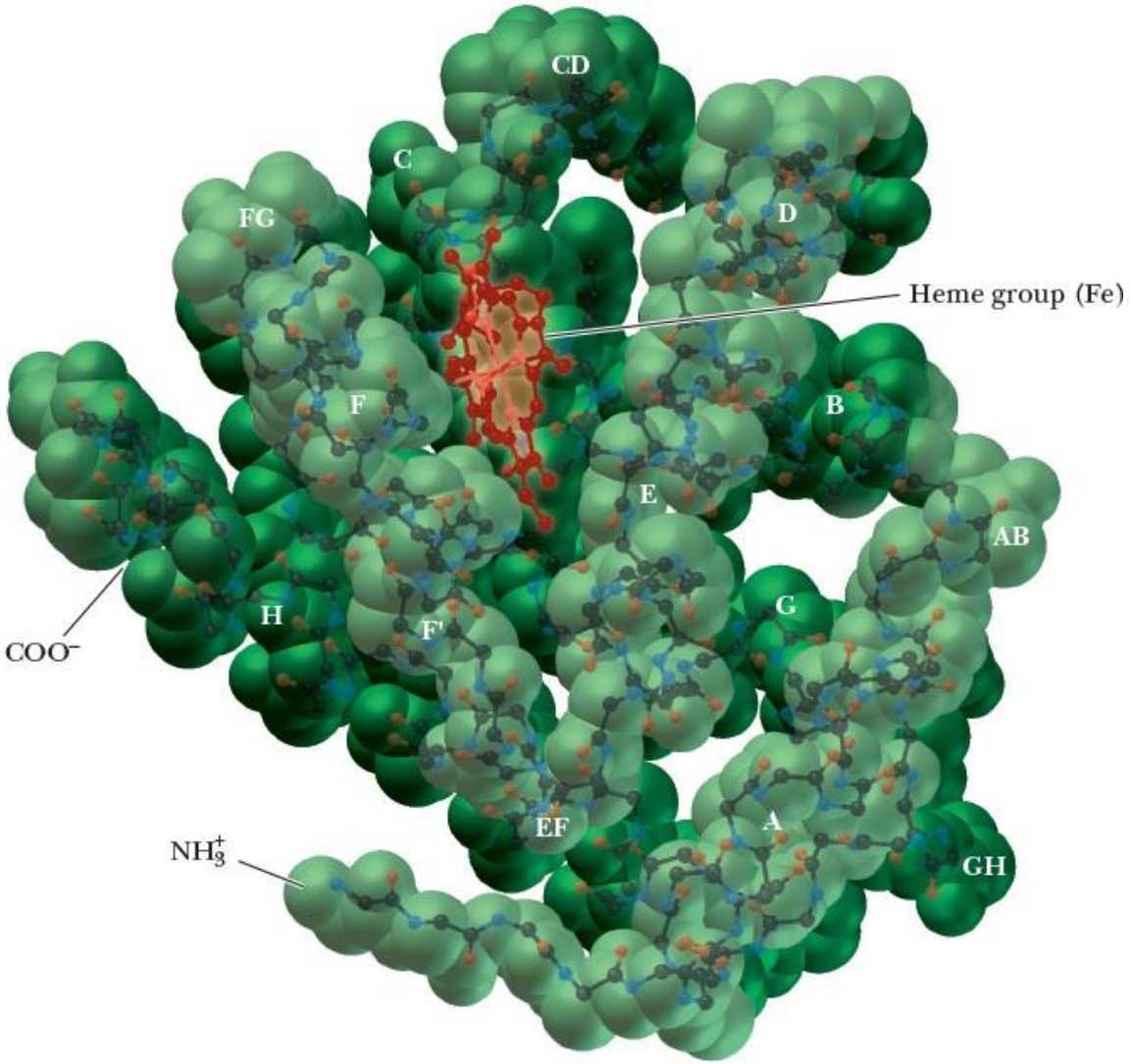


Figure 4.15 - Structure of Myoglobin



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3  What are the noncovalent interactions that occurs between side chains of the following amino acids:

a. Glycine and asparagine

- hydrogen bonding

a. Glutamic acid and lysine

- electrostatic attraction

Quaternary Structure of Proteins

Pertains to proteins that consist of more than one polypeptide chain

- Each chain is called a subunit
- **Oligomers**: Molecules that are made up of a number of smaller subunits
  - Include dimers, trimers, and tetramers
- Chains interact with one another noncovalently via electrostatic attractions, hydrogen bonds, and hydrophobic interactions
  - **Allosteric**: Property of multi-subunit proteins such that a conformational change in one subunit induces a drastic change in another subunit

Quaternary Structure of Proteins

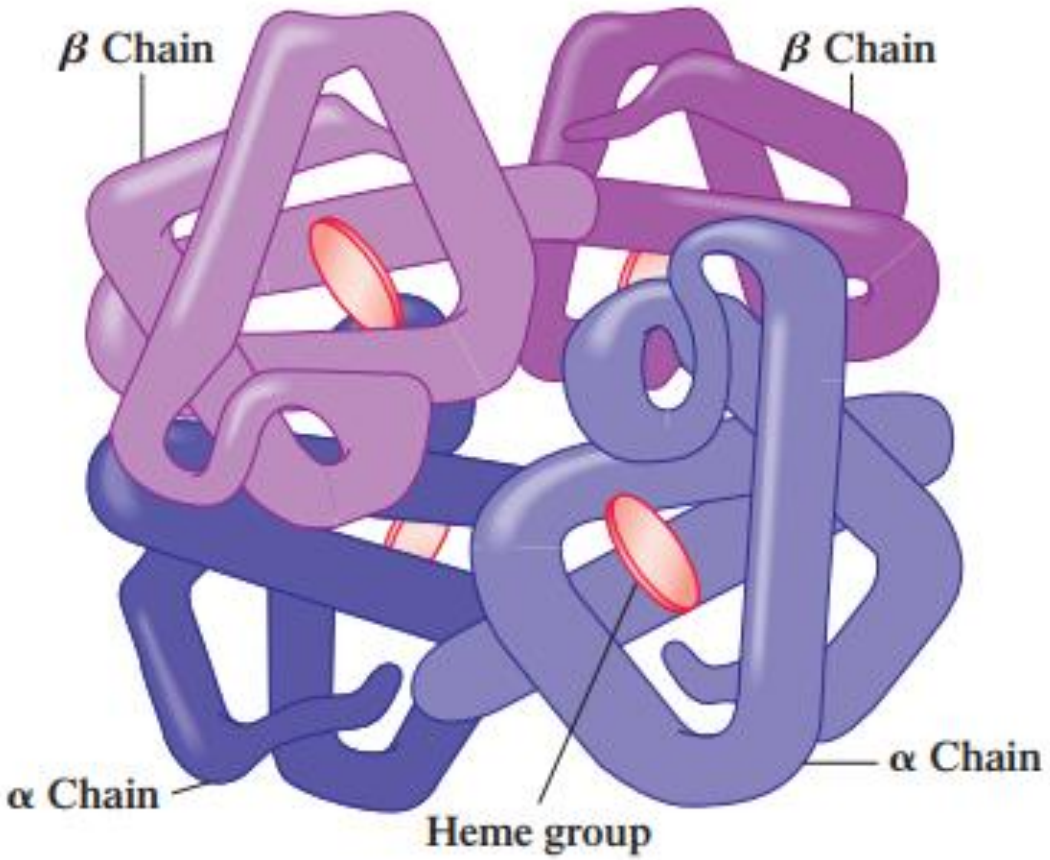


Fig. 23 Tertiary and quaternary structure of the oxygen carrying protein hemoglobin

Tetramer with:

- Two α-chains
  - 141 residues long
- Two β-chains
  - 146 residues long
- Overall structure - $\alpha_2\beta_2$
- Most amino acids of the α-chain, β-chain, and myoglobin are homologous

Hemoglobin

- hemoglobin transports oxygen from the lungs to tissue
- Tetramer (four peptide chains each contain heme group = binds to O_2)
- Iron atom @ center interact w/ O_2 ; 1 hemoglobin: 4 O_2 molecules

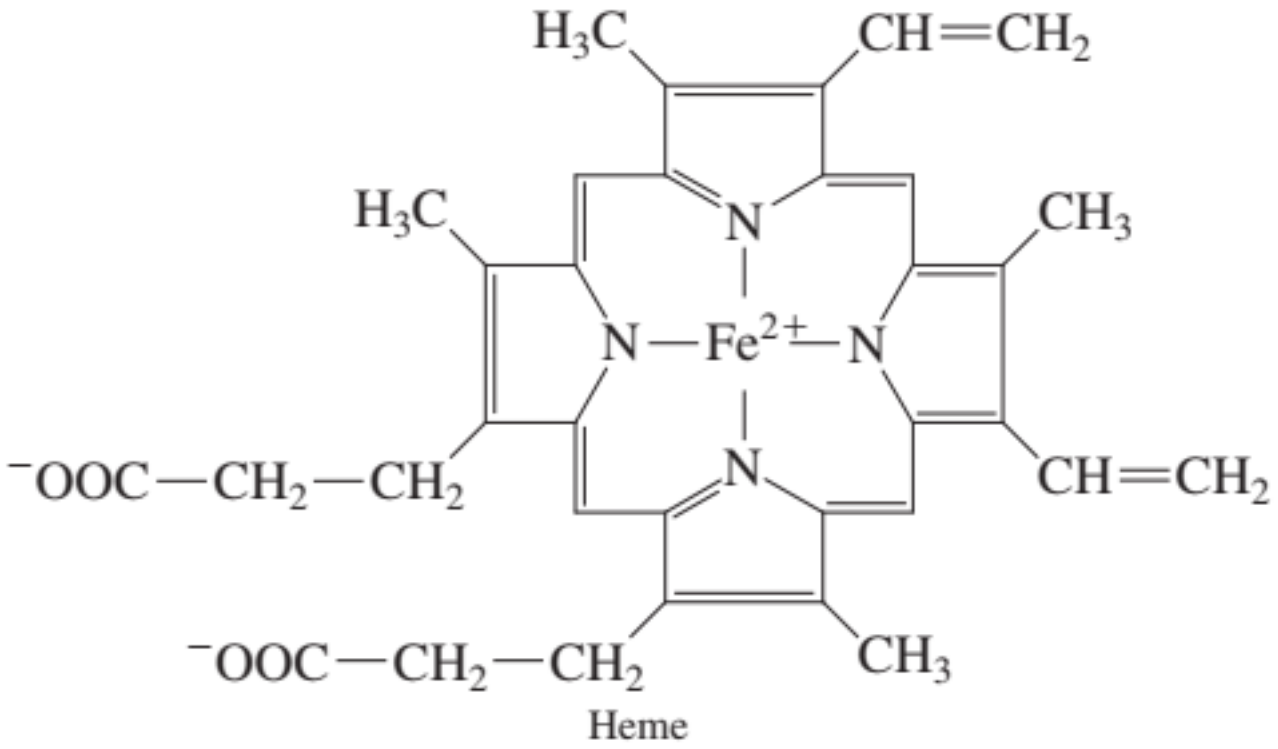


Fig. 31 Structure of heme group

Myoglobin - oxygen storage molecule in muscles (no quaternary structure)

- Single peptide chain & a heme unit hence carry one O_2
- Higher affinity for O_2 than hemoglobin; working muscles

4



Quaternary protein structure is possible for:

a. All proteins

b. Multimeric proteins but not monomeric proteins

c. Monomeric proteins but not multimeric proteins

d. No correct answer

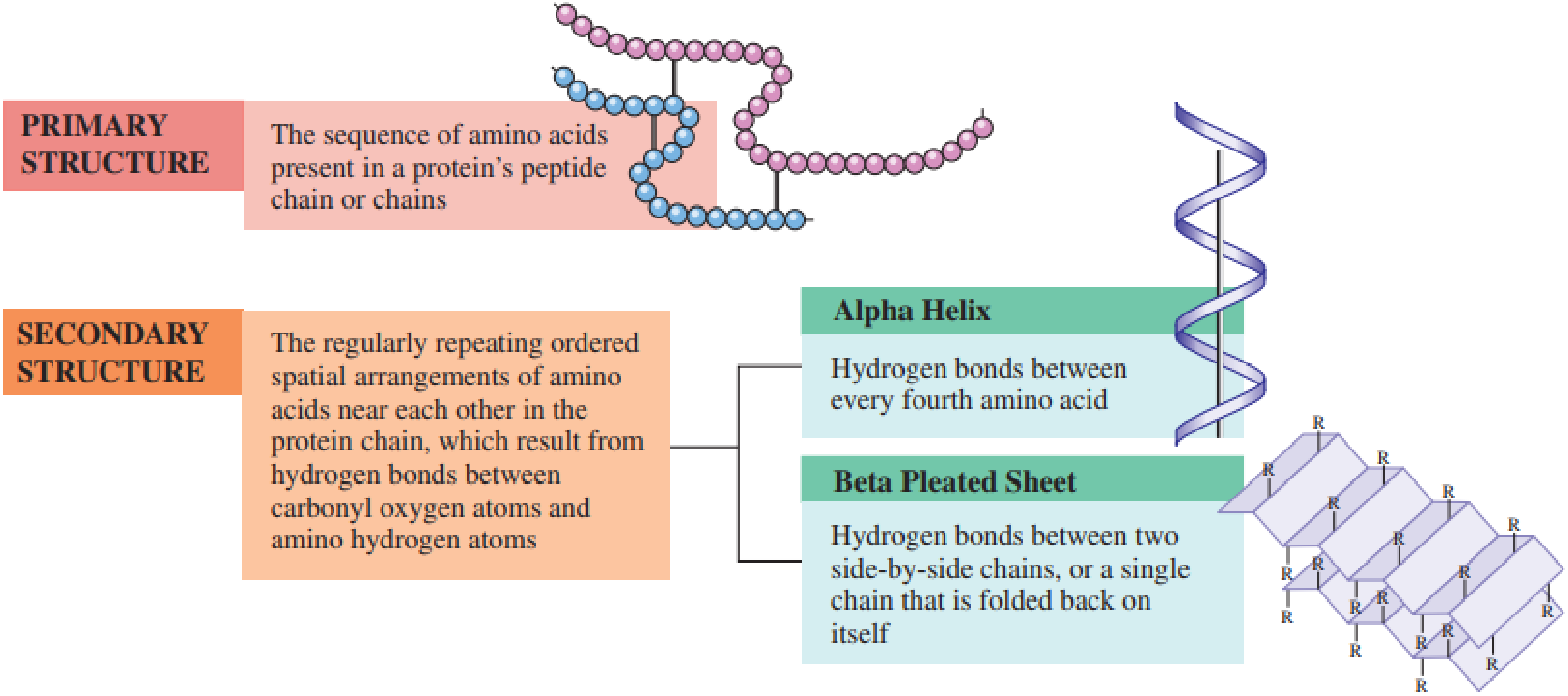
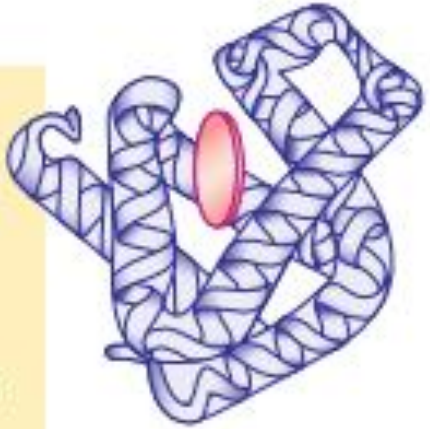


Fig. 24 Summary of primary and secondary structure of proteins

TERTIARY STRUCTURE

The overall three-dimensional shape that results from the attractive forces between amino acid side chains (R groups) that are not near each other in the protein chain

- Disulfide bonds
- Electrostatic interactions
- Hydrogen bonds
- Hydrophobic interactions



QUATERNARY STRUCTURE

The three-dimensional shape of a protein consisting of two or more independent peptide chains, which results from noncovalent interactions between R groups

- Electrostatic interactions
- Hydrogen bonds
- Hydrophobic interactions

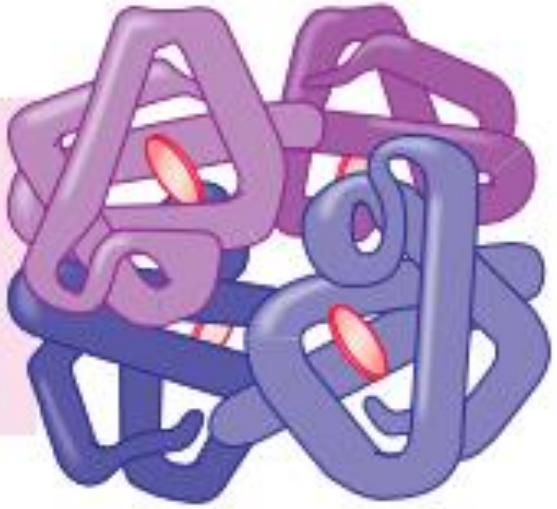


Fig. 25 Summary of tertiary and quaternary structure of proteins

General Structure of Proteins

- Protein - peptide with at least 40 amino acid residues; protein and polypeptide used interchangeably
- 10,000 amino acid on most proteins, 400-500 amino acid residues on some proteins, 40-100 on small proteins.

According to peptide chain

I. Monomeric protein - one peptide chain

II. Multimeric Protein - more than 1 peptide chain

- Peptide chains present = Protein subunits

General Structure of Proteins

According to chemical composition

I. Simple Protein - only amino acids are present, may have subunits.

II. Conjugated Protein - more than one peptide chain and contain *non amino acid* entities

Prosthetic group - non amino acid group present in conjugated protein

- ✓ Lipoproteins - lipid prosthetic groups
- ✓ Glycoproteins - carbohydrate groups
- ✓ Metalloproteins - specific metal, and so on

Table 3. Types of conjugated Proteins

Class	Prosthetic Group	Specific Example	Function of Example
Hemoproteins	Heme Unit	Hemoglobin Myoglobin	Carrier of O2 in blood Oxygen binder in muscle
Lipoproteins	Lipid	LDL HDL	Lipid carrier Lipid carrier
Glycoproteins	Carbohydrates	Gamma globulin Mucin interferon	Antibody Lubricant in mucus Antiviral protection
Phosphoproteins	Phosphate Group	Glycogen phosphorylase	Enzyme in glycogen phosphorylation
Nucleoproteins	Nucleic Acid	Ribosomes Viruses	Site for protein synthesis in cells Self replicating, infectious complex
Metalloproteins	Metal Ion	Iron- ferritin Zinc - alcohol dehydrogenase	Storage for complex ion Enzyme in alcohol oxidation

Classifications of Proteins

According to shape

I. Fibrous – a protein whose molecules w/ elongated shape & one dimension much longer

II. Globular protein – a protein whose molecules have peptide chains that are folded into spherical or globule shapes

Protein Classification Based on Shape

Fibrous protein - molecules w/ elongated shape & one dimension much longer

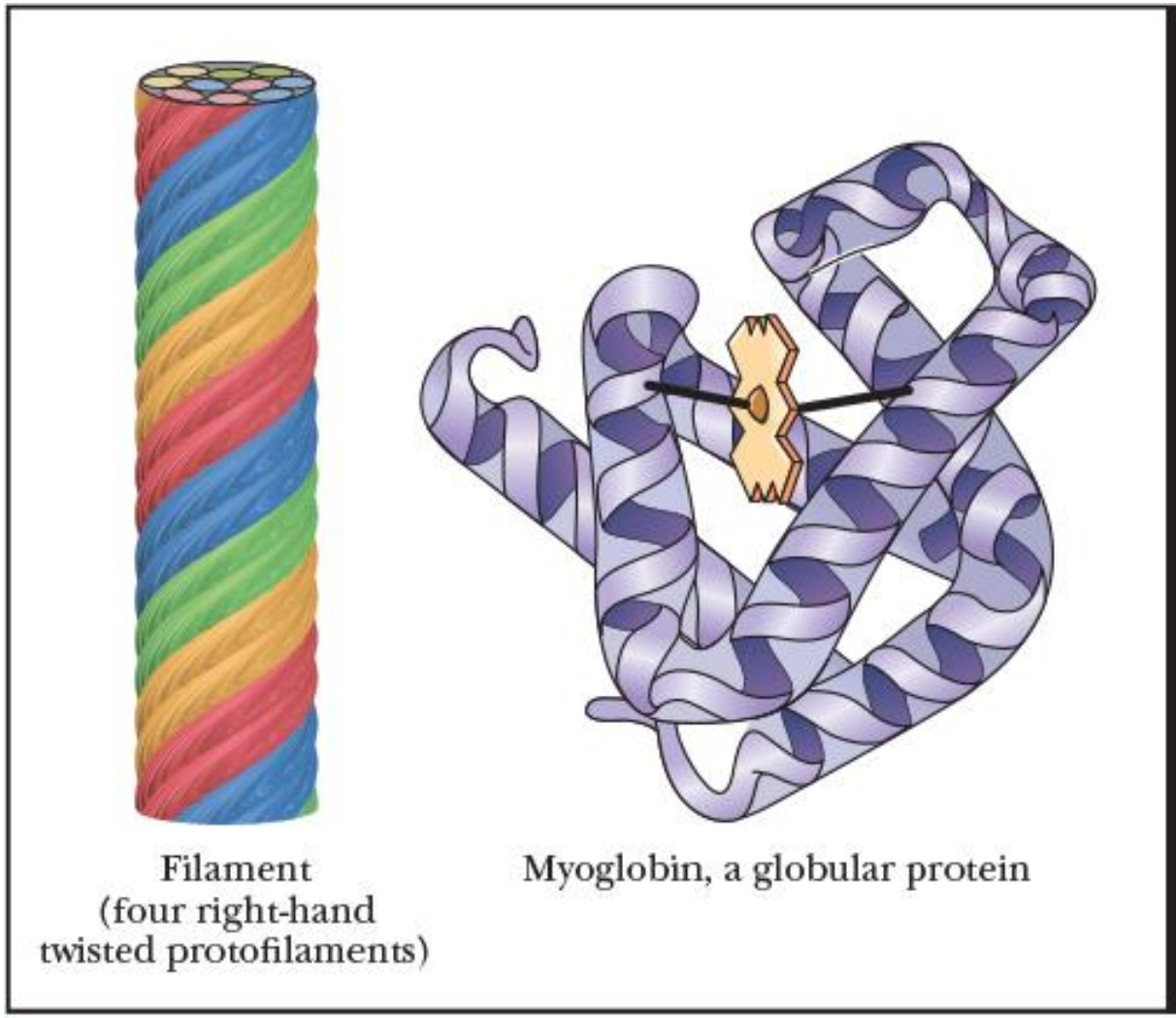
- Linear and form aggregate to form macromolecular structure

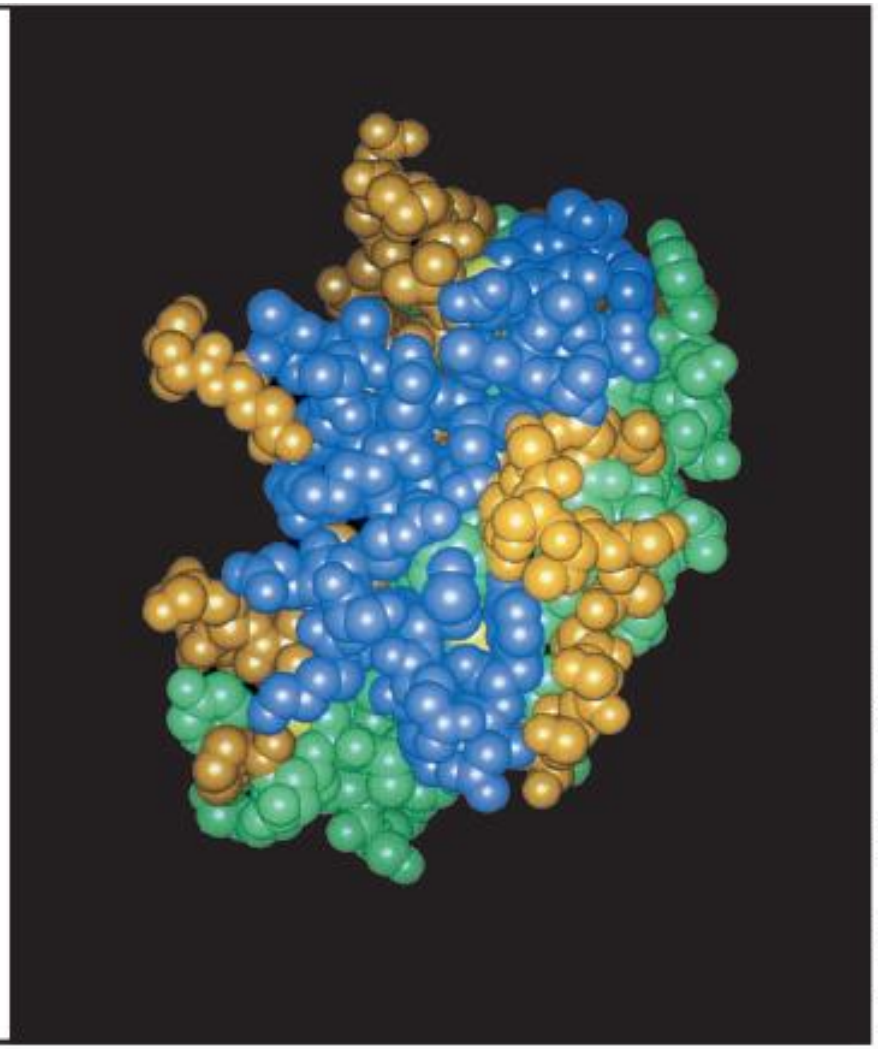
Globular protein - peptide chains folded into spherical or globule shapes

- Hydrophilic side chains on outside; hydrophobic in interior

Characteristic	Fibrous Proteins	Globular Proteins
Shape	Long and elongated (fiber-like)	Compact and spherical (folded)
Solubility	Generally insoluble in water	Generally water-soluble (though some are not)
Secondary Structure	Usually dominated by one repetitive secondary structure (e.g., α-helix, β-sheet, or triple helix)	Often contain a combination of α-helices, β-sheets, and random coils
Main Functions	Structural support, strength, and protection	Diverse biological roles: catalysis (enzymes), transport, regulation (hormones), immunity (antibodies), and metabolism
Examples	Collagen, keratin, elastin, fibroin (spider silk)	Hemoglobin, enzymes (e.g., amylase), insulin, antibodies

Comparison of the Shapes of Fibrous and Globular Proteins



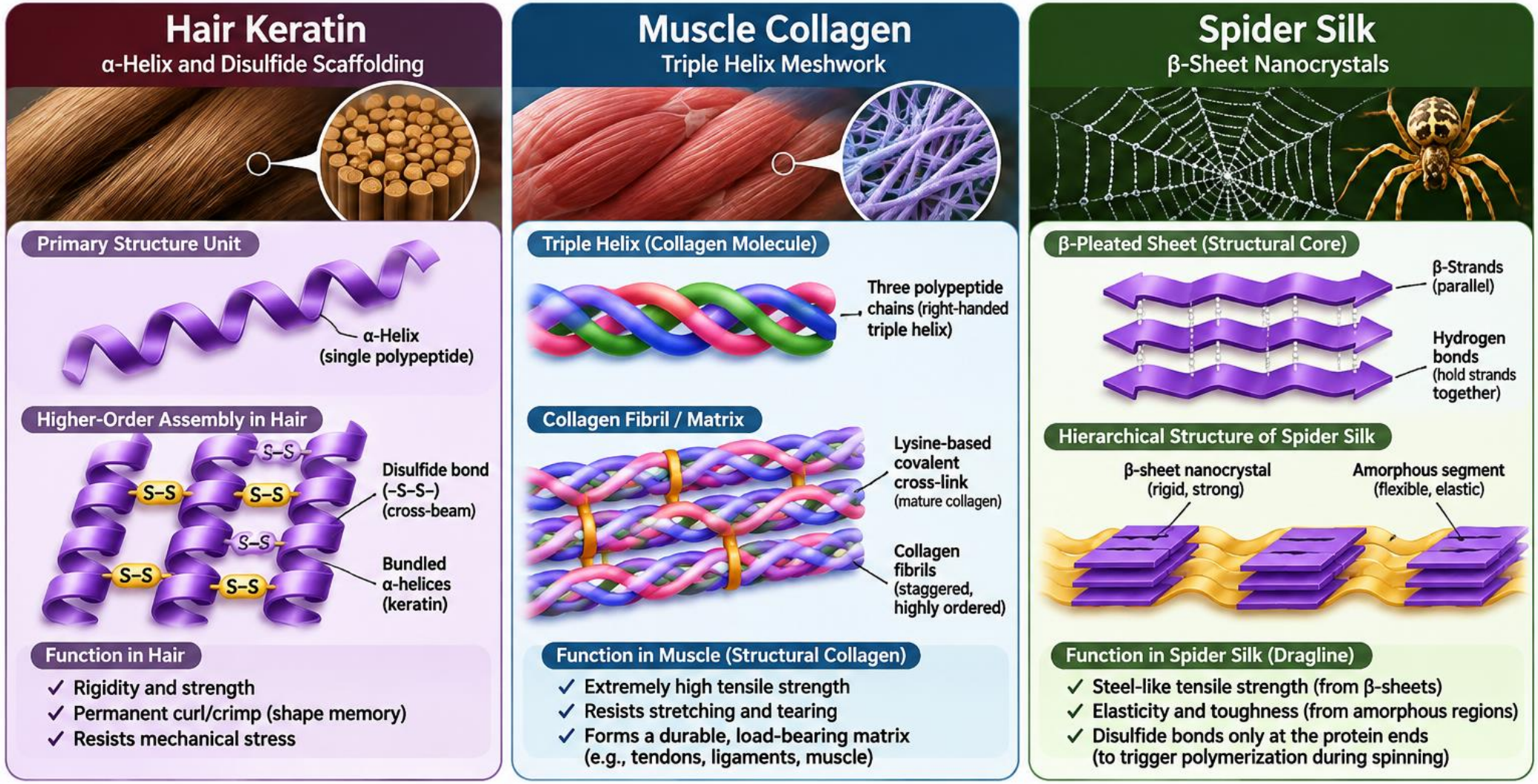


Charles Grisham/University of Virginia

A Schematic diagrams of a portion of a fibrous protein and of a globular protein.

B Computer-generated model of a globular protein. α-helices are shown in blue, β-sheets are green, and random coil is gold.

Some examples of Fibrous Proteins



Feature	Hair Keratin	Muscle Collagen	Spider Silk
Key Folding Structure	α-Helix (coiled)	Triple helix (three chains)	β-Pleated sheets (stacked)
Main Stabilizing Bonds (in mature form)	Disulfide bonds (–S–S–) between helices	Lysine-based covalent cross-links	Dense hydrogen bonds (within β-sheets); rare disulfide bonds at protein ends (for assembly)
Overall Architecture	Bundles of α-helices (coiled–coiled)	Staggered fibrils and fibers (matrix)	β-sheet nanocrystals in a flexible matrix
Resulting Property	Strong, rigid, shape-retaining (hair form, curl)	Very high tensile strength, minimal stretch	Exceptional strength and toughness with elasticity

AI generated image

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